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Machine Learning

Translated by Shaowu Liu

The slides of this chapter
are translated by Lanjihong Ma



Springer

Chapter 9

Clustering

Outlines

- ❑ **Clustering Problem**
- ❑ **Performance Measure**
- ❑ **Distance Calculation**
- ❑ **Prototype Clustering**
- ❑ **Density Clustering**
- ❑ **Hierarchical Clustering**
- ❑ **Probabilistic clustering**

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da
6 CFU

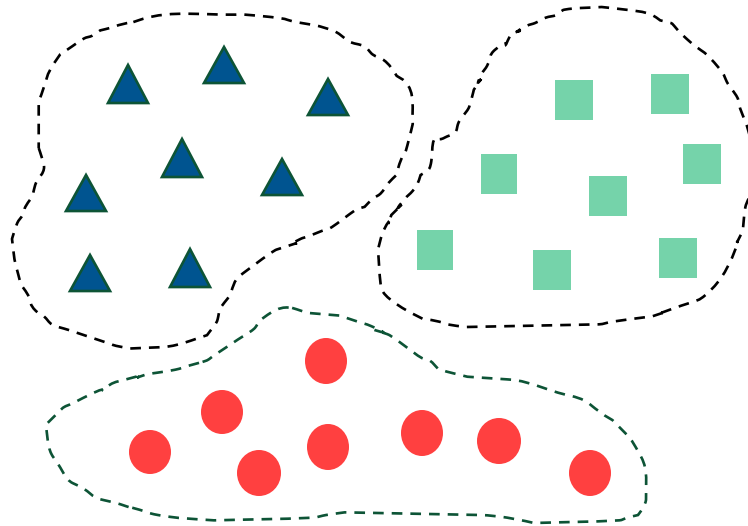
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Outlines

- ❑ **Clustering Problem**
- ❑ Performance Measure
- ❑ Distance Calculation
- ❑ Prototype Clustering
- ❑ Density Clustering
- ❑ Hierarchical Clustering
- ❑ Probabilistic clustering

Clustering Problem

- ❑ It is mostly studied and widely used in *unsupervised learning*.
- ❑ Goal: partitions the data set into several disjoint subsets (clusters).
- ❑ Clustering can be used by itself to identify the inherent structure of data, while it can also serve as a pre-processing technique for other learning tasks such as classification.



Clustering Problem

□ Formalization

Given data set $D = \{x_1, x_2, \dots, x_m\}$ containing m unlabeled samples, where each sample $x_i = (x_{i1}; x_{i2}; \dots; x_{in})$ is a n -dimensional vector.

Then, a clustering algorithm partitions the data set D into k clusters $\{C_l | l = 1, 2, \dots, k\}$, where $C_{l'} \cap_{l' \neq l} C_l = \phi$ and $D = \bigcup_{l=1}^k C_l$.

Accordingly, we denote $\lambda_j \in \{1, 2, \dots, k\}$ as the *cluster label* of sample x_j (i.e., $x_j \in C_{\lambda_j}$). Then the clustering result can be represented as a cluster label vector $\lambda = \{\lambda_1; \lambda_2; \dots; \lambda_m\}$ with m elements.

Outlines

- Clustering Problem
- **Performance Measure**
- Distance Calculation
- Prototype Clustering
- Density Clustering
- Hierarchical Clustering

Performance Measure

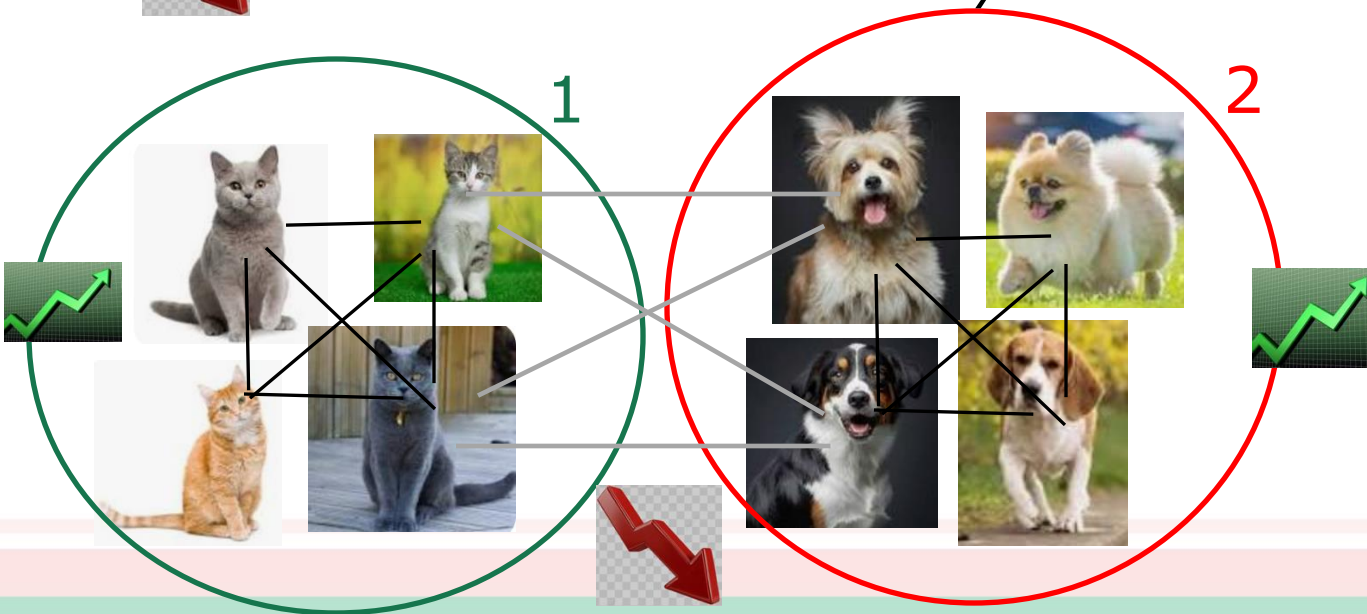
- Performance measure for clustering are also called *validity indices*.
- Intuitively, we wish *things of a kind come together*; that is, samples in the same cluster should be as similar as possible while samples from different clusters should be as different as possible. In other words, we seek clusters with:



high *intra-cluster similarity*



low *inter-cluster similarity*



Performance Measure

□ Performance measure for clustering:

- **External** index

Comparing the clustering result against a *reference model* (labels provided by experts).

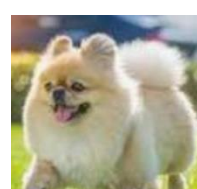
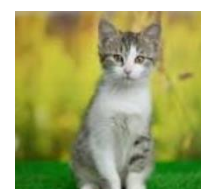
- **Internal** index

Evaluating the clustering result without using any reference model (e.g., variance, cohesion, etc)

Performance Measure–External index

Given a data set $D = \{x_1, x_2, \dots, x_m\}$, suppose a clustering algorithm produces the clusters $C = \{C_1, C_2, \dots, C_k\}$, and a reference model gives the clusters $C^* = \{C_1^*, C_2^*, \dots, C_s^*\}$. Accordingly, let λ and λ^* denote the clustering label vectors of C and C^* .

$$C = \{C_1, C_2\}$$

 x_1  x_2  x_5  x_7  x_3  x_4  x_6  x_8

$$C_1 = \{x_1, x_2, x_5, x_7\}$$

$$C_2 = \{x_3, x_4, x_6, x_8\}$$

$$\lambda = \{1, 1, 2, 2, 1, 2, 1, 2\}$$

Performance Measure–External index

Given a data set $D = \{x_1, x_2, \dots, x_m\}$, suppose a clustering algorithm produces the clusters $C = \{C_1, C_2, \dots, C_k\}$, and a reference model gives the clusters $C^* = \{C_1^*, C_2^*, \dots, C_s^*\}$. Accordingly, let λ and λ^* denote the clustering label vectors of C and C^* .

$$C^* = \{C_1^*, C_2^*\}$$



x_1



x_2



x_5



x_7



x_3



x_4



x_6



x_8

$$C_1^* = \{x_1, x_2, x_3, x_6\}$$

$$C_2^* = \{x_4, x_5, x_7, x_8\}$$

$$\lambda^* = \{1, 1, 1, 2, 2, 1, 2, 2\}$$

Performance Measure–External index

- The clustering algorithm produces the clusters $C = \{C_1, C_2, \dots, C_k\}$
- The reference model gives the clusters $C^* = \{C_1^*, C_2^*, \dots, C_s^*\}$
- λ and λ^* denote the clustering label vectors of C and C^*
- For each pair of samples we can define the following four terms

$$a = |SS|, SS = \{(x_i, x_j) | \lambda_i = \lambda_j, \lambda_i^* = \lambda_j^*, i < j\}$$

$$b = |SD|, SD = \{(x_i, x_j) | \lambda_i = \lambda_j, \lambda_i^* \neq \lambda_j^*, i < j\}$$

$$c = |DS|, DS = \{(x_i, x_j) | \lambda_i \neq \lambda_j, \lambda_i^* = \lambda_j^*, i < j\}$$

$$d = |DD|, DD = \{(x_i, x_j) | \lambda_i \neq \lambda_j, \lambda_i^* \neq \lambda_j^*, i < j\}$$

different
estimated
cluster

same
true
cluster

External index: Example



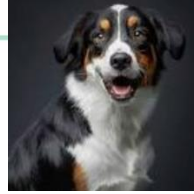
x_1



x_2



x_5



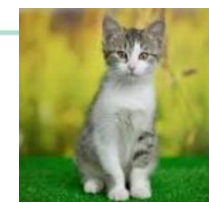
x_7



x_3



x_4



x_6



x_8

$$C_1 = \{x_1, x_2, x_5, x_7\}$$

$$C_2 = \{x_3, x_4, x_6, x_8\}$$

$$C^* = \{C_1^*, C_2^*\} \quad C_1^* = \{x_1, x_2, x_3, x_6\} \quad C_2^* = \{x_4, x_5, x_7, x_8\}$$

$$\lambda = \{\textcolor{red}{1}, \textcolor{red}{1}, \textcolor{yellow}{2}, \textcolor{green}{2}, \textcolor{blue}{1}, \textcolor{yellow}{2}, \textcolor{blue}{1}, \textcolor{green}{2}\}$$

$$\lambda^* = \{\textcolor{red}{1}, \textcolor{red}{1}, \textcolor{yellow}{1}, \textcolor{green}{2}, \textcolor{blue}{2}, \textcolor{yellow}{1}, \textcolor{blue}{2}, \textcolor{green}{2}\}$$

$$a = |SS|, SS = \{(x_i, x_j) | \lambda_i = \lambda_j, \lambda_i^* = \lambda_j^*, i < j\}$$

$$(x_1, x_2), (x_4, x_8), (x_3, x_6), (x_5, x_7)$$

G G C C G G C C

Pairs	$\wedge$ S	$\wedge$ D	Total row
True S	4		
True D			
Total col			28

$m = 8$

$$\frac{m \cdot m - 1}{2}$$

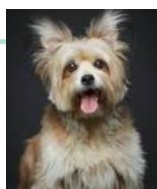
External index: Example



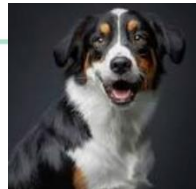
x_1



x_2



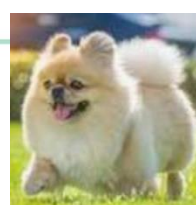
x_5



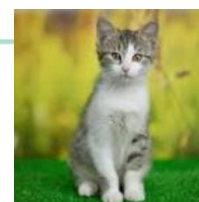
x_7



x_3



x_4



x_6



x_8

$$C_1 = \{x_1, x_2, x_5, x_7\}$$

$$C_2 = \{x_3, x_4, x_6, x_8\}$$

$$C^* = \{C_1^*, C_2^*\} \quad C_1^* = \{x_1, x_2, x_3, x_6\} \quad C_2^* = \{x_4, x_5, x_7, x_8\}$$

$$\lambda = \{1, 1, 2, 2, 1, 2, 1, 2\}$$

$$\lambda^* = \{1, 1, 1, 2, 2, 1, 2, 2\}$$

$$b = |SD|, SD = \{(x_i, x_j) \mid \lambda_i = \lambda_j, \lambda_i^* \neq \lambda_j^*, i < j\}$$

$(x_1, x_5), (x_1, x_7), (x_2, x_5), (x_2, x_7),$

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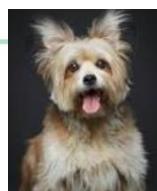
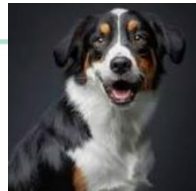
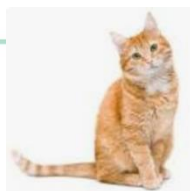
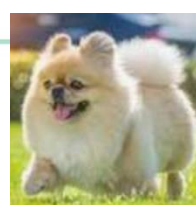
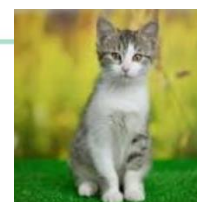
$(x_3, x_4), (x_3, x_8), (x_4, x_6), (x_6, x_8)$

G C G C G C G C

Pairs	$\wedge$ S	$\wedge$ D	Total row
True S	4		
True D	8		
Total col	12	16	28

External index: Example


 x_1

 x_2

 x_5

 x_7

 x_3

 x_4

 x_6

 x_8

$$C_1 = \{x_1, x_2, x_5, x_7\}$$

$$C_2 = \{x_3, x_4, x_6, x_8\}$$

$$C^* = \{C_1^*, C_2^*\} \quad C_1^* = \{x_1, x_2, x_3, x_6\} \quad C_2^* = \{x_4, x_5, x_7, x_8\}$$

$$\lambda = \{1, 1, 2, 2, 1, 2, 1, 2\}$$

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 $(x_1, x_3), (x_1, x_6), (x_2, x_3), (x_2, x_6),$

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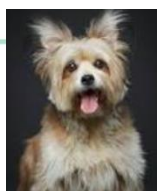
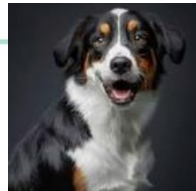
 $(x_4, x_5), (x_4, x_7), (x_5, x_8), (x_7, x_8)$

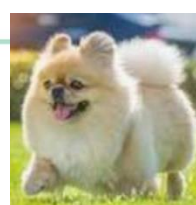
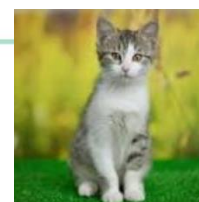
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Pairs	$\wedge$ S	$\wedge$ D	Total row
True S	4	8	12
True D	8		16
Total col	12	16	28

External index: Example


 x_1

 x_2

 x_5

 x_7

 x_3

 x_4

 x_6

 x_8

$$C_1 = \{x_1, x_2, x_5, x_7\}$$

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 $(x_1, x_4), (x_1, x_8), (x_2, x_4), (x_2, x_8),$

C G G C G C G C

 $(x_3, x_5), (x_3, x_7), (x_5, x_6), (x_6, x_7)$

G C G C C G G C

Pairs	$\wedge$ S	$\wedge$ D	Total row
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True D	8	8	16
Total col	12	16	28

Performance Measure–External index

□ Jaccard Coefficient, JC

$$JC = \frac{a}{a+b+c} = \frac{Set_1 \cap Set_2}{Set_1 \cup Set_2}$$

□ Fowlkes and Mallows Index, FMI

(geometric mean) $FMI = \sqrt{\frac{a}{a+b} \cdot \frac{a}{a+c}}$

□ Rand Index, RI

$$RI = \frac{2(a+d)}{m(m-1)}$$

Pairs	$\wedge S$	$\wedge D$	Total row
True S	a	c	a+c
True D	b	d	b+d
Total col	a+b	c+d	a+b+c+d

take values in the interval [0,1], the larger the better

Correctly estimated pairs

Total number of pairs

Performance Measure–External index

□ Jaccard Coefficient, JC

$$JC = \frac{a}{a+b+c} = \frac{4}{20} = 0.2$$

Pairs	$\wedge S$	$\wedge D$	Total row
True S	4	8	12
True D	8	8	16
Total col	12	16	28

□ Fowlkes and Mallows Index, FMI

$$FMI = \sqrt{\frac{a}{a+b} \cdot \frac{a}{a+c}} = \sqrt{\frac{4}{12} \cdot \frac{4}{12}} = \frac{4}{12} = 0.33$$

□ Rand Index, RI

$$RI = \frac{2(a+d)}{m(m-1)} = \frac{12}{28} = 0.428$$

Internal Measures: Cohesion and Separation

- ❑ **Cluster Cohesion**: Measures how closely related are objects in a cluster
 - Example: Sum Squared Error (SSE)
 - Cohesion is measured by the **Within cluster Sum of Squares (WSS)**:

$$WSS = \sum_i \sum_{x \in C_i} (x - m_i)^2$$

The smaller the better.

- ❑ **Cluster Separation**: Measure how distinct or well-separated a cluster is from other clusters
 - Separation is measured by the **Between cluster Sum of Squares (BSS)**:

$$BSS = \sum_i |C_i| (m - m_i)^2$$

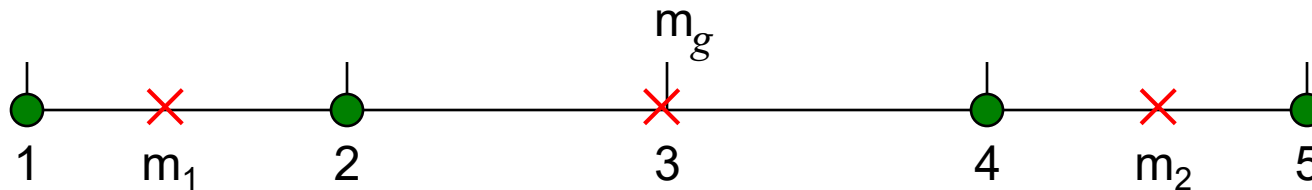
The bigger the better.

- where $|C_i|$ is the size of cluster i , m_i and m_g are the means of cluster i and global mean, respectively

Internal Measures: Cohesion and Separation

□ Example: SSE

- BSS + WSS = constant



K=1 cluster:

$$WSS = (1 - 3)^2 + (2 - 3)^2 + (4 - 3)^2 + (5 - 3)^2 = 10$$

$$BSS = 4 \times (3 - 3)^2 = 0$$

$$Total = 10 + 0 = 10$$

K=2 clusters:

$$WSS = (1 - 1.5)^2 + (2 - 1.5)^2 + (4 - 4.5)^2 + (5 - 4.5)^2 = 1$$

$$BSS = 2 \times (3 - 1.5)^2 + 2 \times (4.5 - 3)^2 = 9$$

$$Total = 1 + 9 = 10$$

R²: a way to combine them

□ $R^2 = 1 - WSS/SSE$

The bigger the better.

- The R-squared index **measures the proportion of the total variance (SSE) in the data that is explained by the clustering solution (WSS),**
- where total variance (SSE) is the variance of the entire dataset,
- and variance within clusters (WSS) is the sum of the variances of each cluster.

Performance Measure–Internal index

□ Given the generated clusters $C = \{C_1, C_2, \dots, C_k\}$, define the following four terms:

- The average distance between the samples in cluster C

$$avg(C) = \frac{2}{|C|(|C|-1)} \sum_{1 \leq i \leq j \leq |C|} dist(x_i, x_j)$$

- The largest distance between samples in cluster C

$$diam(C) = \max_{1 \leq i \leq j \leq |C|} dist(x_i, x_j)$$

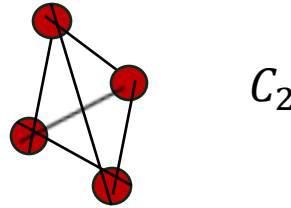
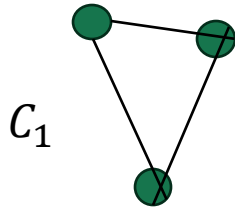
- The distance between two nearest samples in clusters C_i and C_j

$$d_{min}(C) = \min_{x_i \in C_i, x_j \in C_j} dist(x_i, x_j)$$

- The distance between the centroids of clusters C_i and C_j

$$d_{cen}(C) = dist(\mu_i, \mu_j)$$

Performance Measure–Internal index



Number of pairs of
objects in cluster C

$$avg(C) = \frac{2}{|C|(|C|-1)} \sum_{1 \leq i \leq j \leq |C|} dist(x_i, x_j)$$

Sum of the distances
of all the pairs of
objects in cluster C

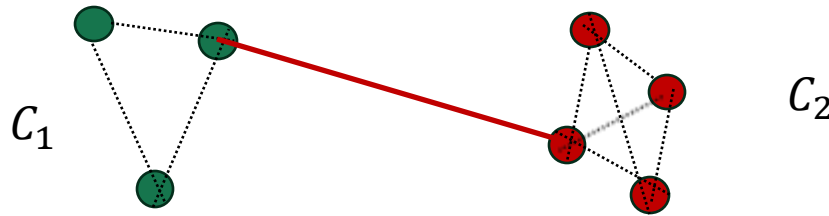
Performance Measure–Internal index



$$avg(C) = \frac{2}{|C|(|C|-1)} \sum_{1 \leq i \leq j \leq |C|} dist(x_i, x_j)$$

$$diam(C) = \max_{1 \leq i \leq j \leq |C|} dist(x_i, x_j)$$

Performance Measure–Internal index

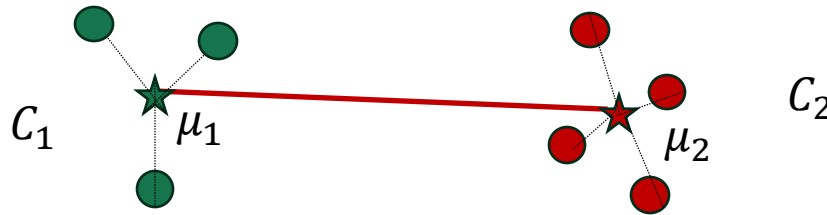


$$avg(C) = \frac{2}{|C|(|C|-1)} \sum_{1 \leq i \leq j \leq |C|} dist(x_i, x_j)$$

$$diam(C) = \max_{1 \leq i \leq j \leq |C|} dist(x_i, x_j)$$

$$d_{min}(C_i, C_j) = \min_{x_i \in C_i, x_j \in C_j} dist(x_i, x_j)$$

Performance Measure–Internal index



$$avg(C) = \frac{2}{|C|(|C|-1)} \sum_{1 \leq i \leq j \leq |C|} dist(x_i, x_j)$$

$$diam(C) = \max_{1 \leq i \leq j \leq |C|} dist(x_i, x_j)$$

$$d_{min}(C_i, C_j) = \min_{x_i \in C_i, x_j \in C_j} dist(x_i, x_j)$$

$$d_{cen}(C_i, C_j) = dist(\mu_i, \mu_j)$$

Performance Measure–Internal index

□ Davies-Bouldin Index, DBI

$$DBI = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{avg(C_i) + avg(C_j)}{d_{cen}(C_i, C_j)} \right)$$

The smaller
the better.

□ Dunn Index, DI

$$DI = \min_{1 \leq i \leq k} \left\{ \min_{j \neq i} \left(\frac{d_{min}(C_i, C_j)}{\max_{1 \leq l \leq k} diam(C_l)} \right) \right\}$$

The bigger
the better.

□ Hausdorff distance

$$Dist_h(X, Z) = \max \{dist_h(X, Z), dist_h(Z, X)\}$$

where

$$dist_h(X, Z) = \max_{x \in X} \min_{z \in Z} \|x - z\|_2$$

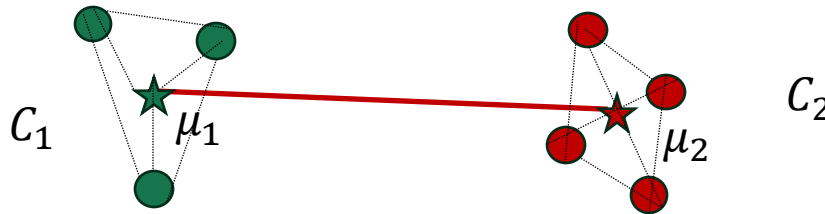
The bigger
the better.

Performance Measure–Internal index

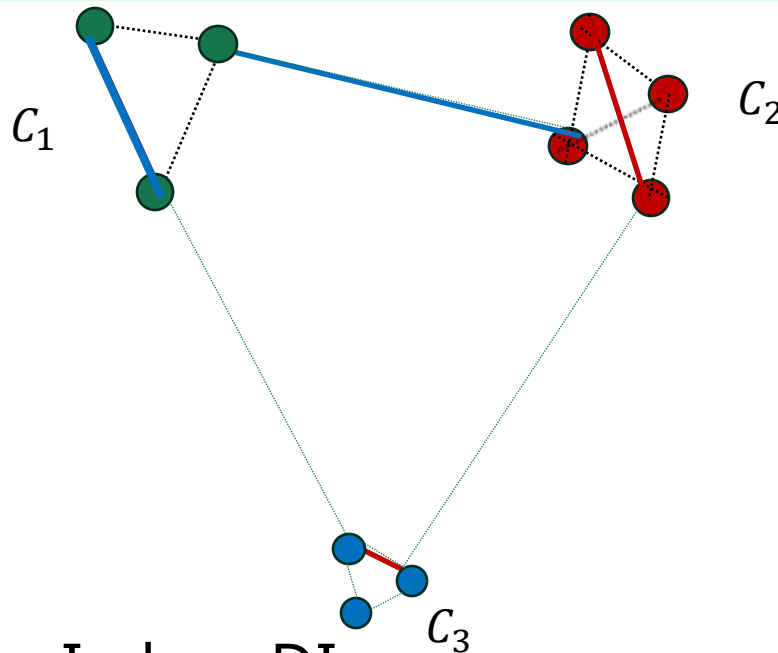
□ Davies-Bouldin Index, DBI

$$DBI = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left(\frac{avg(C_i) + avg(C_j)}{d_{cen}(C_i, C_j)} \right)$$

The smaller
the better.



Performance Measure–Internal index

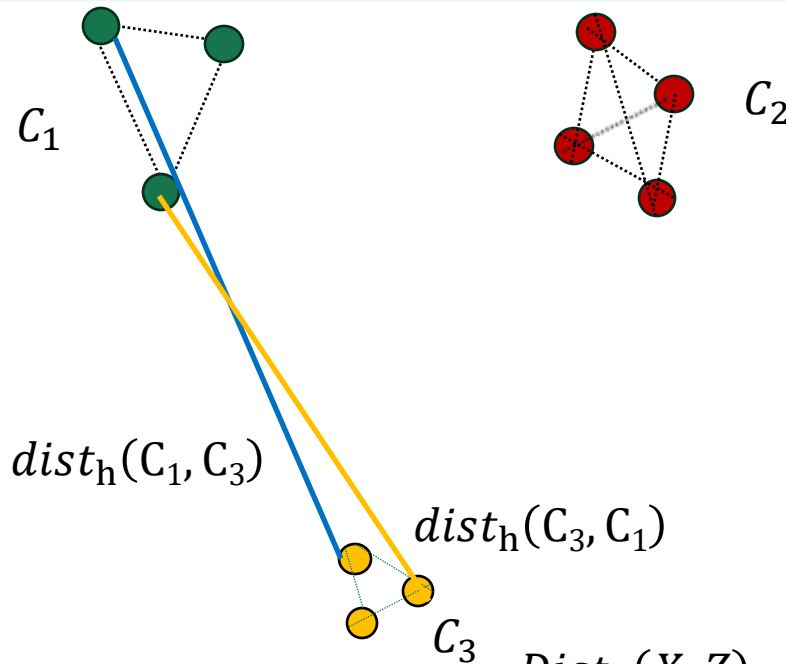


□ Dunn Index, DI

$$DI = \min_{1 \leq i \leq k} \left\{ \min_{j \neq i} \left(\frac{d_{\min}(C_i, C_j)}{\max_{1 \leq l \leq k} \text{diam}(C_l)} \right) \right\}$$

The bigger
the better.

Performance Measure–Internal index



□ Hausdorff distance

(x in X has the most distant nearest neighbor z in Z)

$$Dist_h(X, Z) = \max \{dist_h(X, Z), dist_h(Z, X)\}$$

where

$$dist_h(X, Z) = \max_{x \in X} \min_{z \in Z} \|x - z\|_2$$

A useful equivalence

$$avg(C) = \frac{2}{|C|(|C|-1)} \sum_{1 \leq i \leq j \leq |C|} dist(x_i, x_j)$$

We consider each object pair only once

$$Dis(C) = \frac{1}{|C|^2} \sum_{x_i \in C} \sum_{x_j \in C} [x_i - x_j]^2 = 2 \cdot Var(C)$$

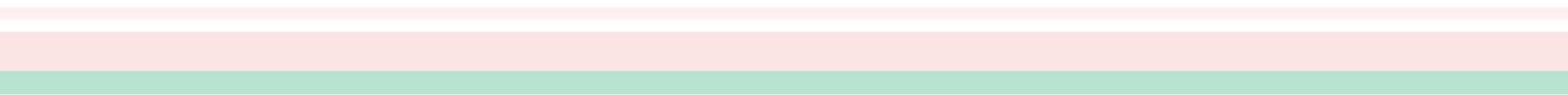
We consider each object pair twice (with objects in a reverse order)

with

$$Var(C) = \frac{1}{|C|} \sum_{x \in C} (x - \bar{x})^2 \quad \bar{x} = \frac{1}{|C|} \sum_{x \in C} x$$

It has been proved that $\bar{x}$ is the location that minimizes the sum of the squared Euclidean distance with the points in a cluster

Outlines

- Clustering Problem
 - Performance Measure
 - **Distance Calculation**
 - Prototype Clustering
 - Density Clustering
 - Hierarchical Clustering
 - Probabilistic clustering
- 

Distance Calculation

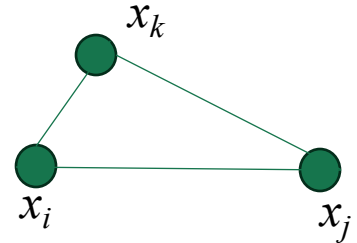
□ The axioms of *distance measure*

Non-negativity: $\text{dist}(x_i, x_j) \geq 0$

Identity of indiscernibles: $\text{dist}(x_i, x_j) = 0$ if and only if $x_i = x_j$

Symmetry: $\text{dist}(x_i, x_j) = \text{dist}(x_j, x_i)$

Subadditivity: $\text{dist}(x_i, x_j) \leq \text{dist}(x_i, x_k) + \text{dist}(x_k, x_j)$



Distance Calculation

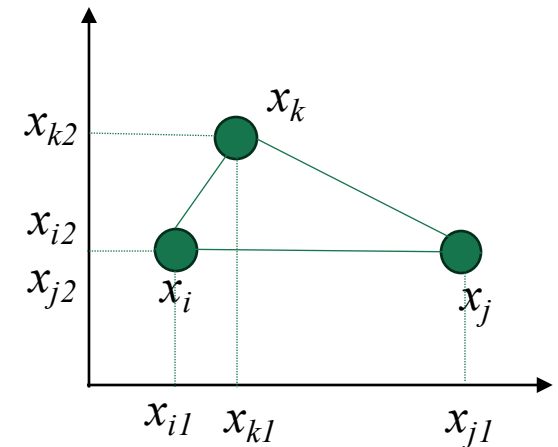
□ The axioms of *distance measure*

Non-negativity: $dist(x_i, x_j) \geq 0$

Identity of indiscernibles: $dist(x_i, x_j) = 0$ if and only if $x_i = x_j$

Symmetry: $dist(x_i, x_j) = dist(x_j, x_i)$

Subadditivity: $dist(x_i, x_j) \leq dist(x_i, x_k) + dist(x_k, x_j)$
(triangular inequality)



Euclidean distance
is a generalization of
Pitagora's theorem

Distance Calculation

□ The axioms of *distance measure*

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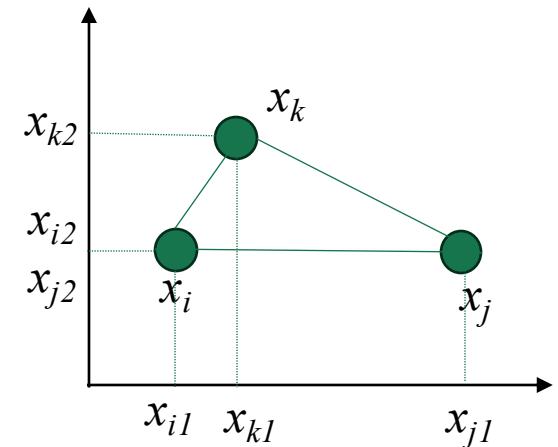
□ A commonly used measure:

Minkowski distance:

$$dist(x_i, x_j) = \left(\sum_{u=1}^n |x_{iu} - x_{ju}|^p \right)^{\frac{1}{p}}$$

p=2: Euclidean distance.

p=1: Manhattan distance.



Euclidean distance
is a generalization of
Pitagora's theorem

Distance Calculation

□ Attributes

- Continuous attribute
having infinite domains
- Categorical attribute
having finite domains

Distance Calculation

□ Attributes

- **Continuous attribute**

having infinite domains (can be manipulated by any Math)

- **Ordinal attribute** (can be ordered or differentiated)

- **Categorical attribute** (can be enumerated or distinguished)
having finite domains

For example, a categorical attribute with the domain $\{1,2,3\}$ is more like an ordinal or continuous attribute since the distance can be calculated with the attribute values, that is, 1 is closer to 2 than 3 .

- **Non-ordinal attribute (or nominal)**

If the domain of an attribute is discrete like $\{\text{aircraft, train, ship}\}$, then the distance cannot be directly calculated with the attribute values, and such attributes are non-ordinal attributes.

Distance Measure

□ Value Difference Metric, VDM:

Let $m_{u,a}$ denote the number of samples taking value a on the attribute u

Let $m_{u,a,i}$ denote the number of samples within the i -th cluster taking value a on the attribute u .

Let k be the number of clusters.

Then the VDM distance between two categorical values a and b of attribute u is

$$VDM_p(a, b) = \sum_{i=1}^k \left| \frac{m_{u,a,i}}{m_{u,a}} - \frac{m_{u,b,i}}{m_{u,b}} \right|^p$$

Example of VDM:

$$VDM_p(a, b) = \sum_{i=1}^k \left| \frac{m_{u,a,i}}{m_{u,a}} - \frac{m_{u,b,i}}{m_{u,b}} \right|^p$$

ID	color	root	sound	texture	umbilicus	surface	ripe (Class)	Cluster
1	green	curly	muffled	clear	hollow	hard	TRUE	1
2	dark	curly	dull	clear	hollow	hard	TRUE	1
3	dark	curly	muffled	clear	hollow	hard	TRUE	1
4	green	curly	dull	clear	hollow	hard	TRUE	1
5	light	curly	muffled	clear	hollow	hard	TRUE	1
6	green	slightly curly	muffled	clear	slightly hollow	soft	TRUE	1
7	dark	slightly curly	muffled	slightly blurry	slightly hollow	soft	TRUE	1
8	dark	slightly curly	muffled	clear	slightly hollow	hard	TRUE	1
9	dark	slightly curly	dull	slightly blurry	slightly hollow	hard	FALSE	2
10	green	straight	crisp	clear	flat	soft	FALSE	2
11	light	straight	dull	blurry	flat	hard	FALSE	2
12	light	curly	muffled	blurry	flat	soft	FALSE	2
13	green	slightly curly	muffled	slightly blurry	hollow	hard	FALSE	2
14	light	slightly curly	dull	slightly blurry	hollow	hard	FALSE	2
15	dark	slightly curly	muffled	clear	slightly hollow	soft	FALSE	2
16	light	curly	muffled	blurry	flat	hard	FALSE	2
17	green	curly	dull	slightly blurry	slightly hollow	hard	FALSE	2

$$VDM_2(\text{green}, \text{dark}) = \left| \frac{3}{6} - \frac{4}{6} \right|^2 + \left| \frac{3}{6} - \frac{2}{6} \right|^2 = 0.055$$

$$VDM_2(\text{dark}, \text{light}) = \left| \frac{4}{6} - \frac{1}{5} \right|^2 + \left| \frac{2}{6} - \frac{4}{5} \right|^2 = 0.435$$

$$VDM_2(\text{green}, \text{light}) = \left| \frac{3}{6} - \frac{1}{5} \right|^2 + \left| \frac{3}{6} - \frac{4}{5} \right|^2 = 0.18$$

Distance Measure with mixed attribute types

□ MinkovDM_p:

N of continuous attributes

Categorical attributes

$$MinkovDM_p(\vec{x}_i, \vec{x}_j) = \left(\sum_{u=1}^{n_c} |x_{iu} - x_{ju}|^p + \sum_{u=n_c+1}^n VDM_p(x_{iu}, x_{ju}) \right)^{\frac{1}{p}}$$

□ Weighted distance

- When different attributes have different importance:

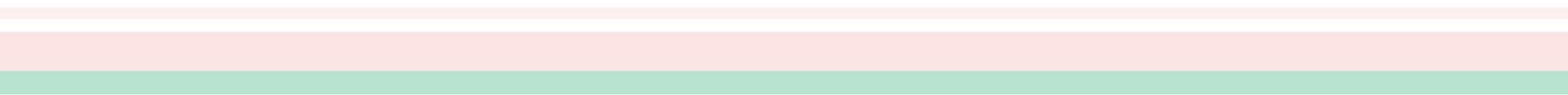
$$dist(\vec{x}_i, \vec{x}_j) = (\omega_1 \cdot |x_{i1} - x_{j1}|^p + \dots + \omega_n \cdot |x_{in} - x_{jn}|^p)^{\frac{1}{p}}$$

Distance using attribute l

$\omega_i \geq 0, \sum_{i=1}^n \omega_i = 1$

Distance using attribute n

Outlines

- Clustering Problem
 - Performance Measure
 - Distance Calculation
 - **Prototype Clustering**
 - Density Clustering
 - Hierarchical Clustering
 - Probabilistic clustering
- 

Prototype Clustering

□ Prototype Clustering

Also known as *prototype-based clustering*, assumes the clustering structure can be represented with set of prototypes.

□ Algorithm:

Typically, such algorithms start with some initial prototypes, and then iteratively update and optimize the prototypes.

□ Next we discuss several well-known prototype-based clustering algorithms.

K-means Clustering, Learning Vector Quantization, Mixture-of-Gaussian Clustering

Prototype Clustering – *k*-means Clustering

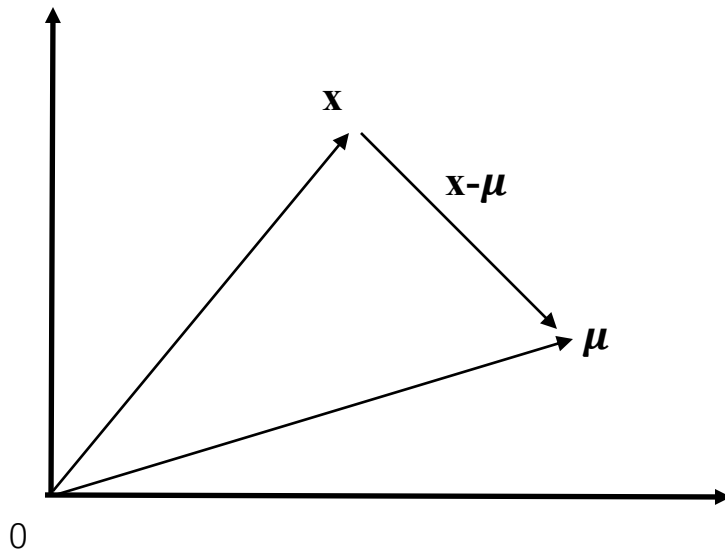
Given a data set $D = \{x_1, x_2, \dots, x_m\}$, the *k*-means algorithm minimizes the squared error of clusters $C = \{C_1, C_2, \dots, C_k\}$:

$$E = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|_2^2$$

where μ_i is the mean vector of cluster C_i (denoted also by $\bar{x}$).

Intuitively, E represents the closeness between the mean vector of a cluster and the samples within that cluster, where a smaller E indicates higher intra-cluster similarity.

The norm-2 of a vector



$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix}$$

$$x - \mu = \begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \\ \vdots \\ x_n - \mu_n \end{bmatrix}$$

$$\|x - \mu\|_2 = \sqrt{(x_1 - \mu_1)^2 + (x_2 - \mu_2)^2 + \cdots + (x_n - \mu_n)^2}$$

$$\|x - \mu\|_2^2 = (x_1 - \mu_1)^2 + (x_2 - \mu_2)^2 + \cdots + (x_n - \mu_n)^2 = \sum_{i=1}^n (x_i - \mu_i)^2$$

$$\|x - \mu\|_2^2 = (X - \mu)^T (X - \mu) \quad \text{in vector notation}$$

Prototype Clustering – *k*-means Clustering

- Given a data set $D = \{x_1, x_2, \dots, x_m\}$, the *k*-means algorithm minimizes the squared error of clusters $C = \{C_1, C_2, \dots, C_k\}$:

$$E = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|_2^2$$

μ_i is the mean vector of cluster C_i

- Algorithm (iteratively optimization)
 - initializes the mean vectors of clusters**
 - repeat**
 1. update the clusters
 2. calculate the mean vectors
 - until** clusters do not change

Prototype Clustering – *k*-means Clustering

□ Pseudocode:

Algorithm 9.1 *k*-means Clustering.

Input: Data set $D = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$;
Number of clusters k .

Process:

```
1: Randomly select  $k$  samples as the initial mean vectors  $\{\mu_1, \mu_2, \dots, \mu_k\}$ ;  
2: repeat  
3:    $C_i = \emptyset (1 \leq i \leq k)$ ;  
4:   for  $j = 1, 2, \dots, m$  do  
5:     Compute the distance between sample  $\mathbf{x}_j$  and each mean vector  $\mu_i (1 \leq i \leq k)$ :  
        $d_{ji} = \|\mathbf{x}_j - \mu_i\|_2$ ;  
6:     According to the nearest mean vector, decide the cluster label of  $\mathbf{x}_j$ :  $\lambda_j =$   
        $\arg \min_{i \in \{1, 2, \dots, k\}} d_{ji}$ ;  
7:     Move  $\mathbf{x}_j$  to the corresponding cluster:  $C_{\lambda_j} = C_{\lambda_j} \cup \{\mathbf{x}_j\}$ ;  
8:   end for  
9:   for  $i = 1, 2, \dots, k$  do  
10:    Compute the updated mean vectors:  $\mu'_i = \frac{1}{|C_i|} \sum_{\mathbf{x} \in C_i} \mathbf{x}$ ;  
11:    if  $\mu'_i \neq \mu_i$  then  
12:      Update the current mean vector  $\mu_i$  to  $\mu'_i$ ;  
13:    else  
14:      Leave the current mean vector unchanged.  
15:    end if  
16:  end for  
17: until All mean vectors remain unchanged  
Output: Clusters  $\mathcal{C} = \{C_1, C_2, \dots, C_k\}$ .
```

Prototype Clustering – *k*-means Clustering

□ An example for *k*-means algorithm

We take the watermelon data set 4:0 in the following table as an example to demonstrate the *k*-means algorithm. For ease of discussion, let x_i represent the sample with the ID i

ID	density	sugar	ID	density	sugar	ID	density	sugar
1	0.697	0.460	11	0.245	0.057	21	0.748	0.232
2	0.774	0.376	12	0.343	0.099	22	0.714	0.346
3	0.634	0.264	13	0.639	0.161	23	0.483	0.312
4	0.608	0.318	14	0.657	0.198	24	0.478	0.437
5	0.556	0.215	15	0.360	0.370	25	0.525	0.369
6	0.403	0.237	16	0.593	0.042	26	0.751	0.489
7	0.481	0.149	17	0.719	0.103	27	0.532	0.472
8	0.437	0.211	18	0.359	0.188	28	0.473	0.376
9	0.666	0.091	19	0.339	0.241	29	0.725	0.445
10	0.243	0.267	20	0.282	0.257	30	0.446	0.459

Prototype Clustering – *k*-means Clustering

□ An example for *k*-means algorithm

Suppose we set $k=3$, then the algorithm randomly picks up three samples x_6, x_{12}, x_{24} as the initial mean vectors, that is, $\mu_1 = (0.403; 0.237)$, $\mu_2 = (0.343; 0.099)$, $\mu_3 = (0.478; 0.437)$

Then, for the sample $x_1 = (0.697; 0.460)$, its distances to the three current mean vectors μ_1, μ_2, μ_3 are 0.369, 0.506, and 0.220, respectively. Thus x_1 is assigned to cluster C_3 . Similarly, we evaluate all samples in the data set and find the following cluster assignments:

$$C_1 = \{x_3, x_5, x_6, x_7, x_8, x_9, x_{10}, x_{13}, x_{14}, x_{17}, x_{18}, x_{19}, x_{20}, x_{23}\}$$

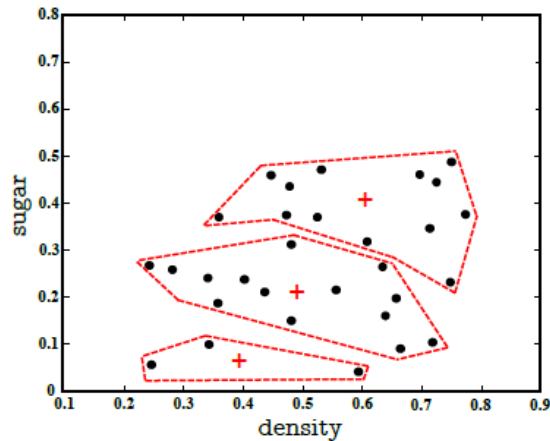
$$C_2 = \{x_{11}, x_{12}, x_{16}\}$$

$$C_3 = \{x_1, x_2, x_4, x_{15}, x_{21}, x_{22}, x_{24}, x_{25}, x_{26}, x_{27}, x_{28}, x_{29}, x_{30}\}$$

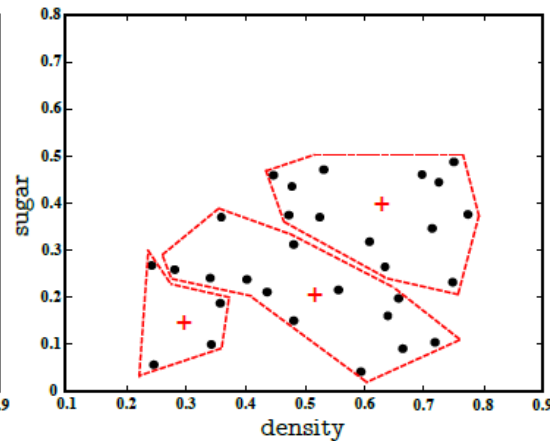
$$\mu'_1 = (0.493; 0.207), \mu'_2 = (0.394; 0.066), \mu'_3 = (0.602; 0.396)$$

Prototype Clustering – *k*-means Clustering

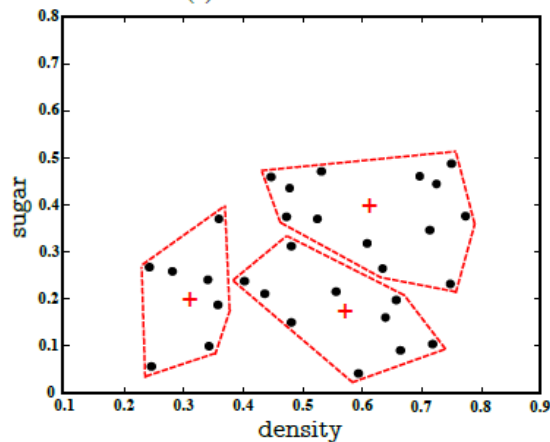
□ Results of the *k*-means algorithm



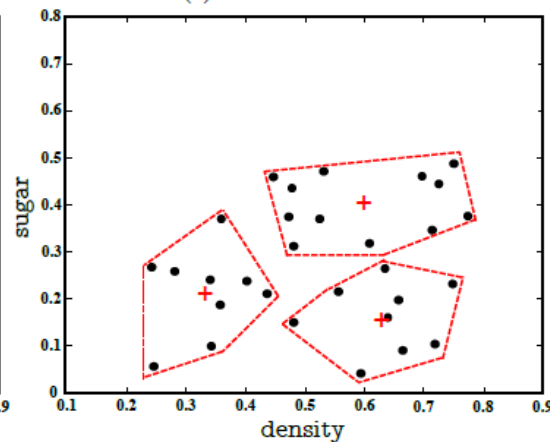
(a) After 1 iteration.



(b) After 2 iterations.



(c) After 3 iterations.



(d) After 4 iterations.

Prototype Clustering – Learning Vector Quantization

□ Learning Vector Quantization, LVQ

Unlike typical clustering algorithm, LVQ assumes data samples are labeled, and the clustering process is assisted by supervised information.

Given a data set $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$,

LVQ aims to learn a set of n -dimensional prototype vectors $\{p_1, p_2, \dots, p_q\}$, where each prototype vector represents one cluster.

□ Quantization is the process of reducing the precision of a signal or set of values by mapping a larger (continuous) set of inputs to a smaller, discrete set of outputs. This technique is used in various fields, such as digital signal processing and machine learning, to make data more efficient by reducing memory and computation requirements, albeit at the cost of some loss of accuracy.

Prototype Clustering – Learning Vector Quantization

□ Pseudocode:

Algorithm 9.2 Learning Vector Quantization.

Input: Training set $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$;

Number of prototype vectors q ;

Initial labels of prototype vectors $\{t_1, t_2, \dots, t_q\}$;

Learning rate η .

Process:

1: Initialize a set of prototype vectors $\{p_1, p_2, \dots, p_q\}$;

2: **repeat**

3: Randomly pickup a sample (x_j, y_j) from the data set D ;

4: Compute the distance between x_j and p_i ($1 \leq i \leq q$): $d_{ji} = \|x_j - p_i\|_2$;

5: Find the nearest prototype vector p_{i^*} for x_j , where $i^* = \arg \min_{i \in \{1, 2, \dots, q\}} d_{ji}$;

6: **if** $y_j = t_{i^*}$ **then**

7: $p' = p_{i^*} + \eta \cdot (x_j - p_{i^*})$; // p is moved toward x_j

8: **else** (their distance is decreased by η)

9: $p' = p_{i^*} - \eta \cdot (x_j - p_{i^*})$; // p is moved away from x_j

10: **end if** (their distance is increased by η)

11: Update the prototype vector p_{i^*} to p' .

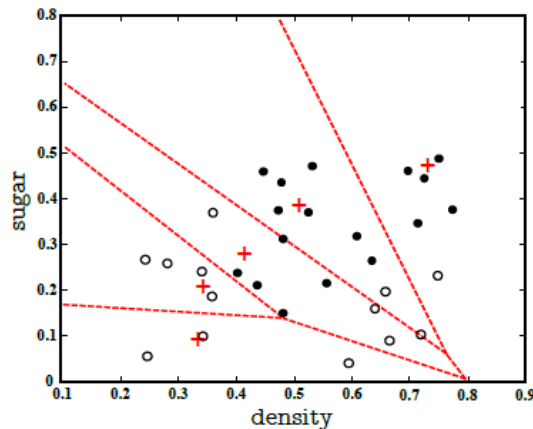
12: **until** The termination condition is met

Output: Prototype vectors $\{p_1, p_2, \dots, p_q\}$.

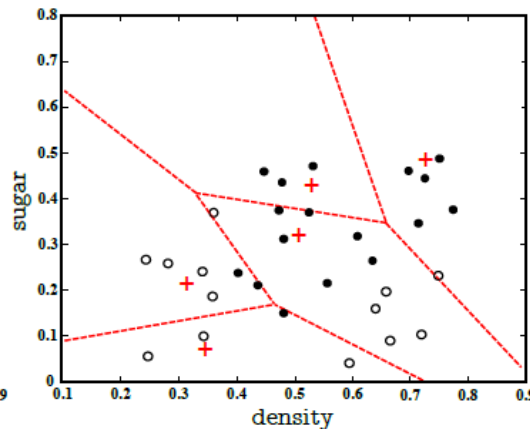
$$\begin{aligned}\|p' - x_j\|_2 &= \|p_{i^*} + \eta \cdot (x_j - p_{i^*}) - x_j\|_2 \\ &= (1 - \eta) \cdot \|p_{i^*} - x_j\|_2.\end{aligned}$$

Prototype Clustering – Learning Vector Quantization

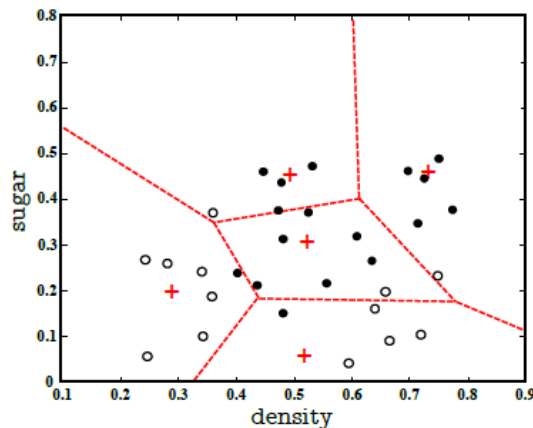
□ Clustering results



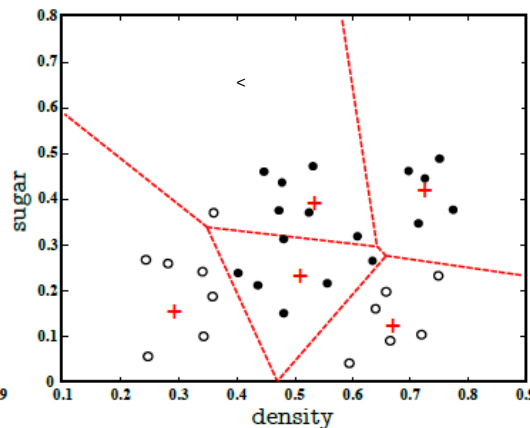
(a) After 50 iterations.



(b) After 100 iterations.



(c) After 200 iterations.

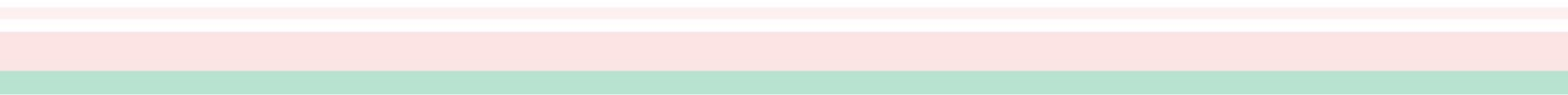


(d) After 400 iterations.

A prototype vector p_i defines a region R_i , where the distance from any sample in it to p_i is not larger than the distance to any other prototype vector. The regions form a *Voronoi tessellation*

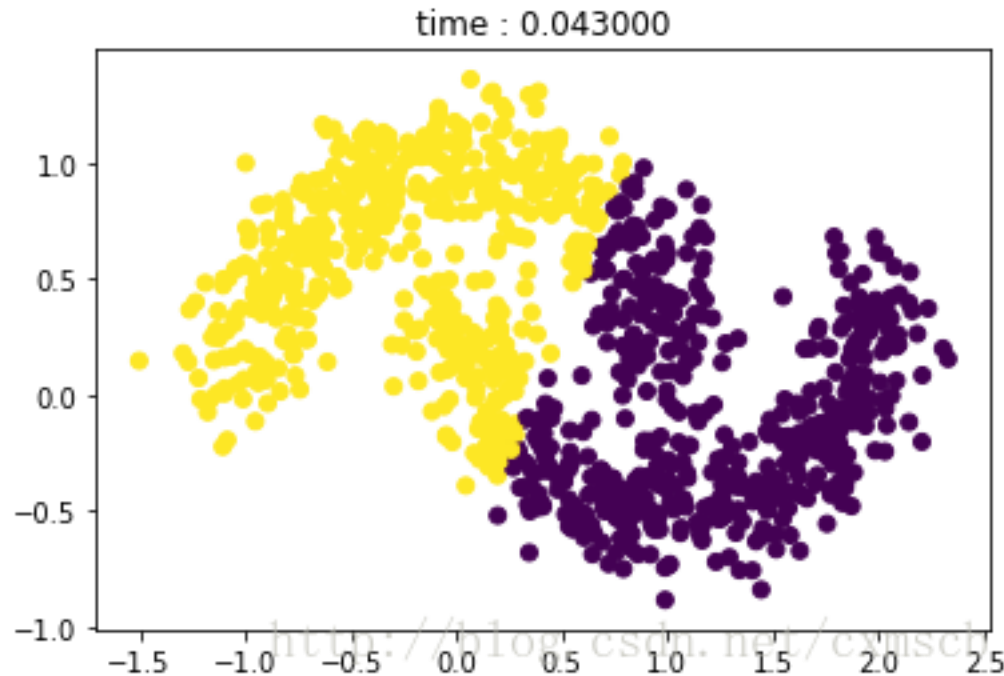
If all samples are represented by the prototype vectors, then we have done a *lossy compression* to the data: this is why it is called vector quantization.

Outlines

- Clustering Problem
 - Performance Measure
 - Distance Calculation
 - Prototype Clustering
 - **Density Clustering**
 - Hierarchical Clustering
 - Probabilistic clustering
- 

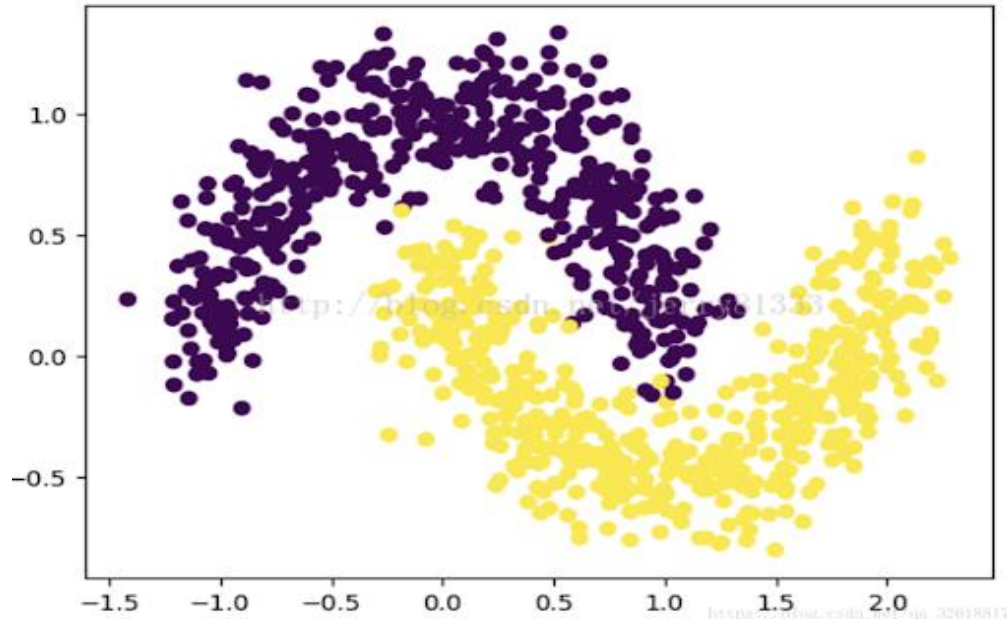
Density Clustering

Applying k -means clustering to following data, it is hard for prototype-based clustering to find the density information, which leads to a further results from expected:



Density Clustering

Density-based clustering: evaluate the connectivity between samples from the density perspective and expand the clusters by adding connectable samples.



Density Clustering

□ Definition of Density Clustering

Density Clustering is also known as *density-based clustering*.

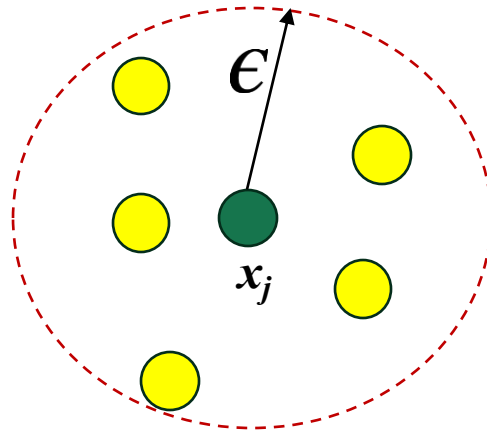
Assuming the clustering structure can be determined by the densities of sample distributions.

Typically, density clustering algorithms evaluate the connectivity between samples from the density perspective.
It expands the clusters by adding connectable samples.

We introduce DBSCAN, which is a representative density clustering algorithm.

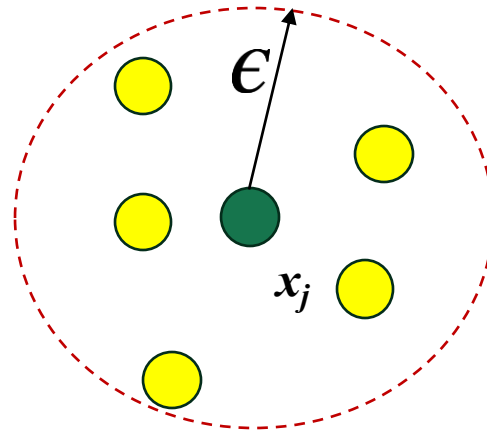
Density Clustering

- DBSCAN characterizes the density of sample distributions by a pair of *neighborhood* parameters $(\epsilon, MinPts)$
- Define the basic concepts:
 - **ϵ -neighborhood**: for $x_j \in D$, its ϵ -neighborhood includes all samples in D that have a distance to x_j no larger than ϵ ;



Density Clustering

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- Define the basic concepts:
 - **ϵ -neighborhood**: for $x_j \in D$, its ϵ -neighborhood includes all samples in D that have a distance to x_j no larger than ϵ ;
 - **Core object**: if the ϵ -neighborhood of x_j includes at least $MinPts$ samples, then x_j is called a core object;

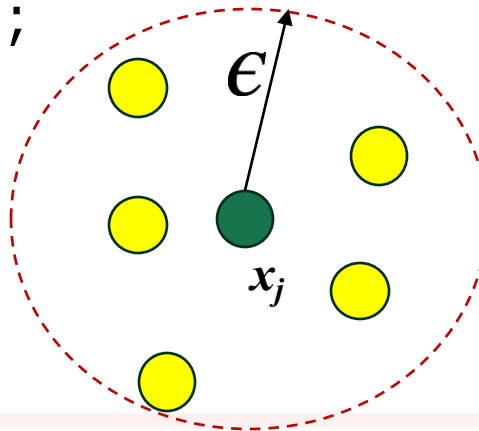


$$\text{density}(x_j) = 6$$

If $MinPts \leq 6 \rightarrow x_j$ is core point

Density Clustering

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 - **Directly density-reachable**: x_j is said to be directly density-reachable by x_i if x_i is a core object and x_j is in the ϵ -neighborhood of x_i ;



MinPts = 4

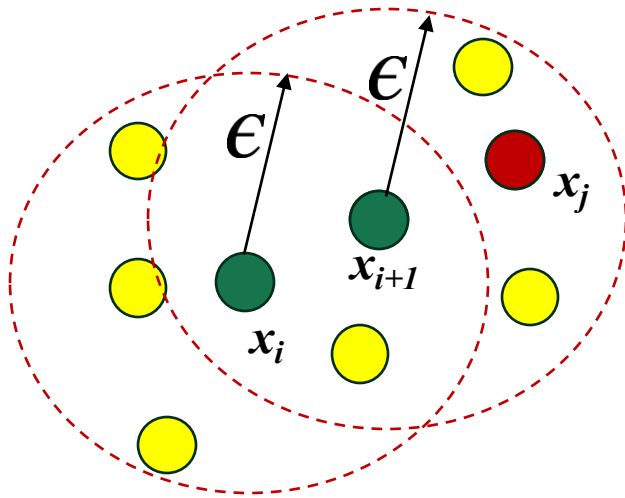
All ● points are
density reachable
by x_j

Density Clustering

- ❑ DBSCAN characterizes the density of sample distributions by a pair of *neighborhood* parameters $(\epsilon, MinPts)$
- ❑ Define the basic concepts:
 - **ϵ -neighborhood**: for $x_j \in D$, its ϵ -neighborhood includes all samples in D that have a distance to x_j no larger than ϵ ;
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 - **Directly density-reachable**: x_j is said to be directly density-reachable by x_i if x_i is a core object and x_j is in the ϵ -neighborhood of x_i ;
 - **Density-reachable**: x_j is said to be density-reachable by x_i if there exists a sequence of samples $p_1, p_2, \dots, p_n$, where $p_1 = x_i, p_n = x_j$ and p_{i+1} is directly density-reachable by p_i ;

Example on Density Reachable

x_j is density reachable by x_i by means of x_{i+1}

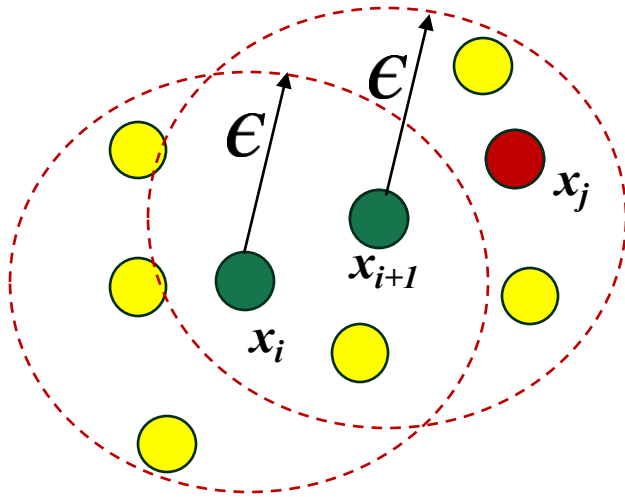


Density Clustering

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- Define the basic concepts:
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 - **Density-reachable**: x_j is said to be density-reachable by x_i if there exists a sequence of samples $p_1, p_2, \dots, p_n$, where $p_1 = x_i, p_n = x_j$ and p_{i+1} is directly density-reachable by p_i ;
 - **Density-connected**: x_i and x_j are density-connected if there exists x_k such that both are density-reachable by x_k

Example on Density Connected

x_j and x_i are density connected because there exists x_{i+1}

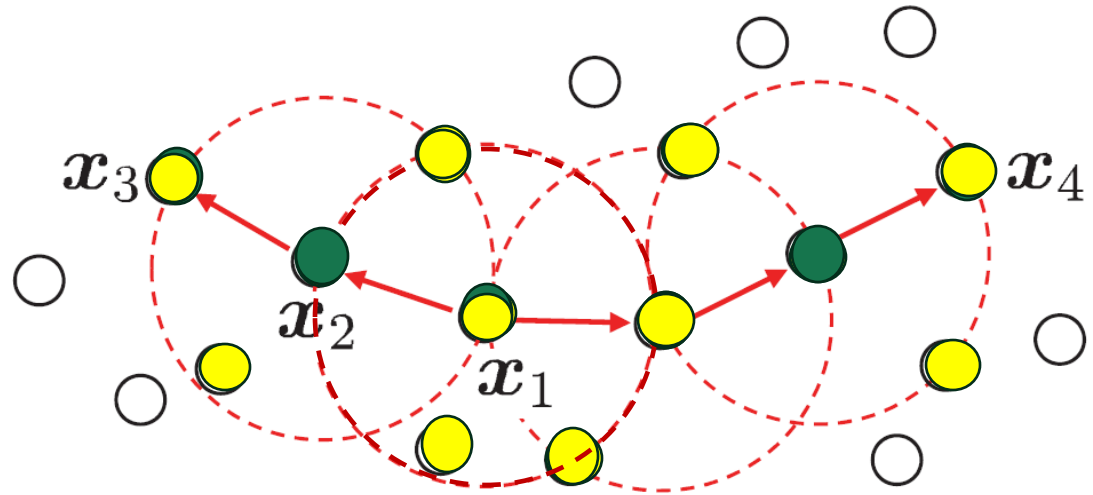


Density Clustering

□ An example

Let $MinPts = 3$:

the dashed circles show
the ϵ -neighborhood



x_1 is a core object

x_2 is directly density-reachable by x_1

x_3 is density-reachable by x_1

x_3 and x_4 density-connected.

Density Clustering

□ Definition of a cluster

The *largest set* of density-connected samples derived by density-reachable relationships.

□ Formalization

Given the neighborhood parameters, a cluster is a non-empty subset with following properties:

Connectivity: $x_i \in C, x_j \in C \Rightarrow x_i$ and x_j are density-connected

Maximality: $x_i \in C, x_j$ is density-reachable by $x_i \Rightarrow x_j \in C$

Actually:

1. if x is a core object and
2. let $X = \{x' \in D \mid x' \text{ is density-reachable by } x\}$ denote the set of samples density-reachable by x ,

then it can be proved that X **is a cluster that satisfies both the connectivity and the maximality.**

Density Clustering

□ Pseudocode for DBSCAN:

Algorithm 9.4 DBSCAN.

Input: Data set $D = \{x_1, x_2, \dots, x_m\}$;
Neighborhood parameters $(\epsilon, MinPts)$.

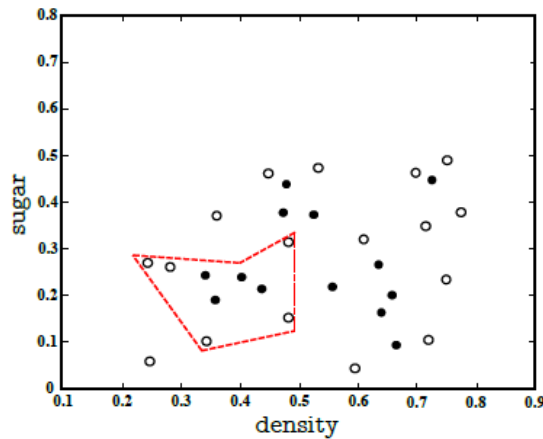
Process:

```
1: Initialize the set of core objects:  $\Omega = \emptyset$ ;  
2: for  $j = 1, 2, \dots, m$  do  
3:   Determine the  $\epsilon$ -neighborhood  $N_\epsilon(x_j)$  of sample  $x_j$ ;  
4:   if  $|N_\epsilon(x_j)| \geq MinPts$  then  
5:     Add sample  $x_j$  to the set of core objects:  $\Omega = \Omega \cup \{x_j\}$ ;  
6:   end if  
7: end for  
8: Initialize the number of clusters:  $k = 0$ ;  
9: Initialize the set of unprocessed samples:  $\Gamma = D$ ;  
10: while  $\Omega \neq \emptyset$  do  
11:   Keep a copy of the current unprocessed data set:  $\Gamma_{old} = \Gamma$ ;  
12:   Randomly select a core object  $o \in \Omega$ , and initialize the queue  $Q = \langle o \rangle$ ;  
13:    $\Gamma = \Gamma \setminus \{o\}$ ;  
14:   while  $Q \neq \emptyset$  do  
15:     Dequeue the first sample  $q$  from  $Q$ ;  
16:     if  $|N_\epsilon(q)| \geq MinPts$  then  
17:       Letting  $\Delta = N_\epsilon(q) \cap \Gamma$ ; //  $\Delta$  has the new objects to be added to the new cluster  
18:       Enqueue the samples in  $\Delta$  into  $Q$ ;  
19:        $\Gamma = \Gamma \setminus \Delta$ ;   $\Gamma$  contains examples not in the  
20:     end if next cluster  
21:   end while  
22:    $k = k + 1$ , generate cluster  $C_k = \Gamma_{old} \setminus \Gamma$ ;   $C_k$  by a double subtraction  
23:    $\Omega = \Omega \setminus C_k$ . satisfies the maximality  
24: end while
```

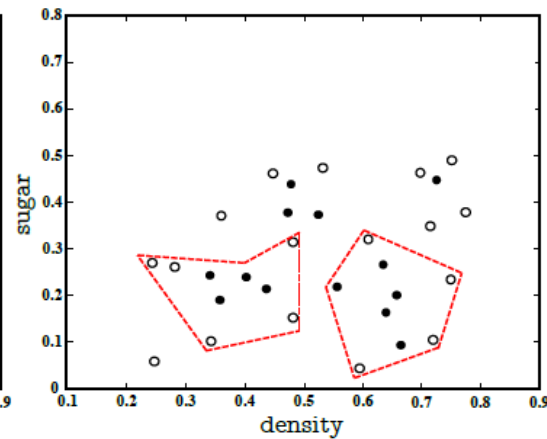
Output: Clusters $\mathcal{C} = \{C_1, C_2, \dots, C_k\}$.

Density Clustering

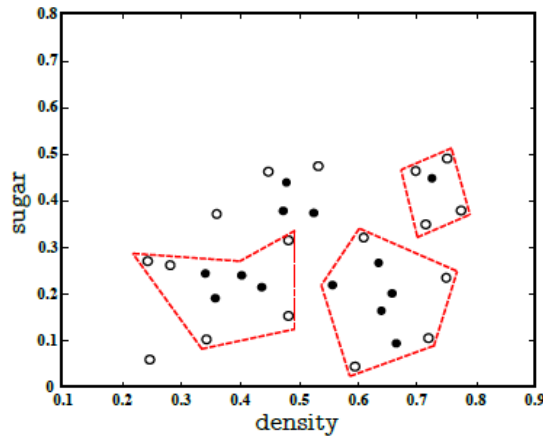
□ Clustering results:



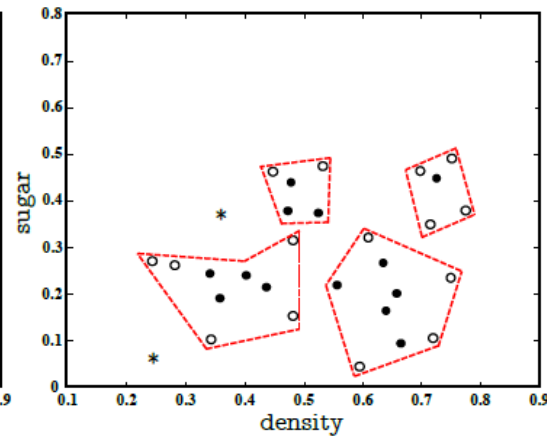
(a) Generating cluster C_1 .



(b) Generating cluster C_2 .



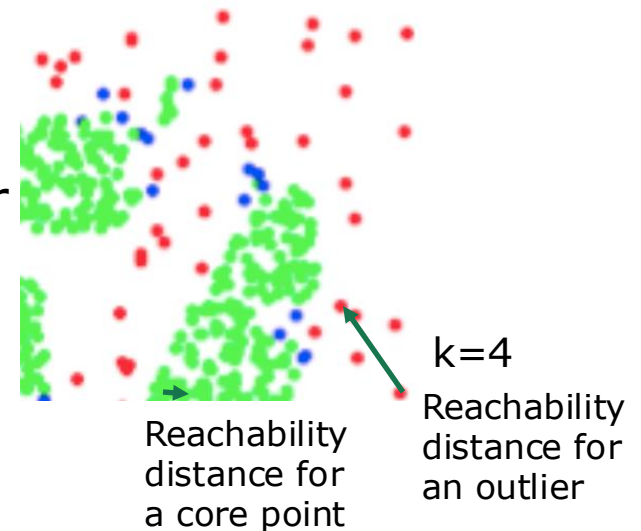
(c) Generating cluster C_3 .



(d) Generating cluster C_4 .

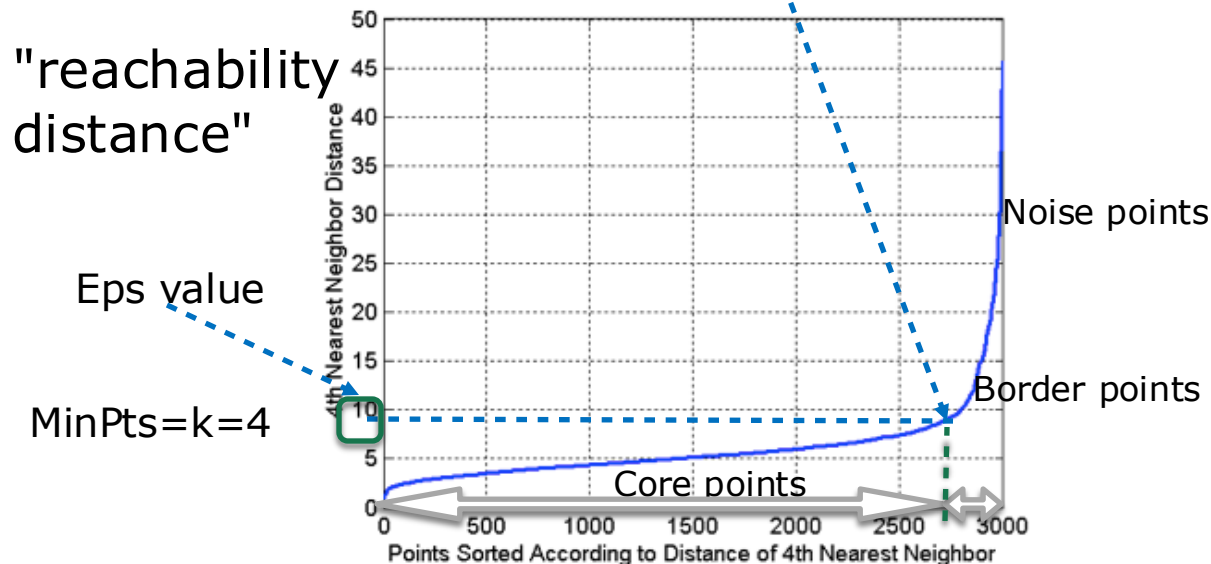
How to compute DBSCAN parameters

- ❑ Clusters are areas of the instance space with the same, high density
- ❑ The density of examples in the neighbourhood of each example is verified by computing the distance with its k -th nearest neighbour. Let us call it "reachability distance". We set k equal to a constant value that is set as the MinPts parameter.
- ❑ if the point is a core point, the density in its neighbourhood is high (the point is surrounded by other close points) and the reachability distance is low
- ❑ If the point is a noise point, it is an outlier (the point is isolated) and its reachability distance is high



How to compute DBSCAN parameters

- ❑ The reachability distance is computed for every point
- ❑ The dataset is sorted in an ascending way for reachability distance
- ❑ In the following plot each point is represented with its order in the abscissa, and its reachability distance on the y axis
- ❑ When the plot shows a "knee" that is the right value of the distance at which we should set the Eps parameter value and discriminate between core and noise (and border) points



Outlines

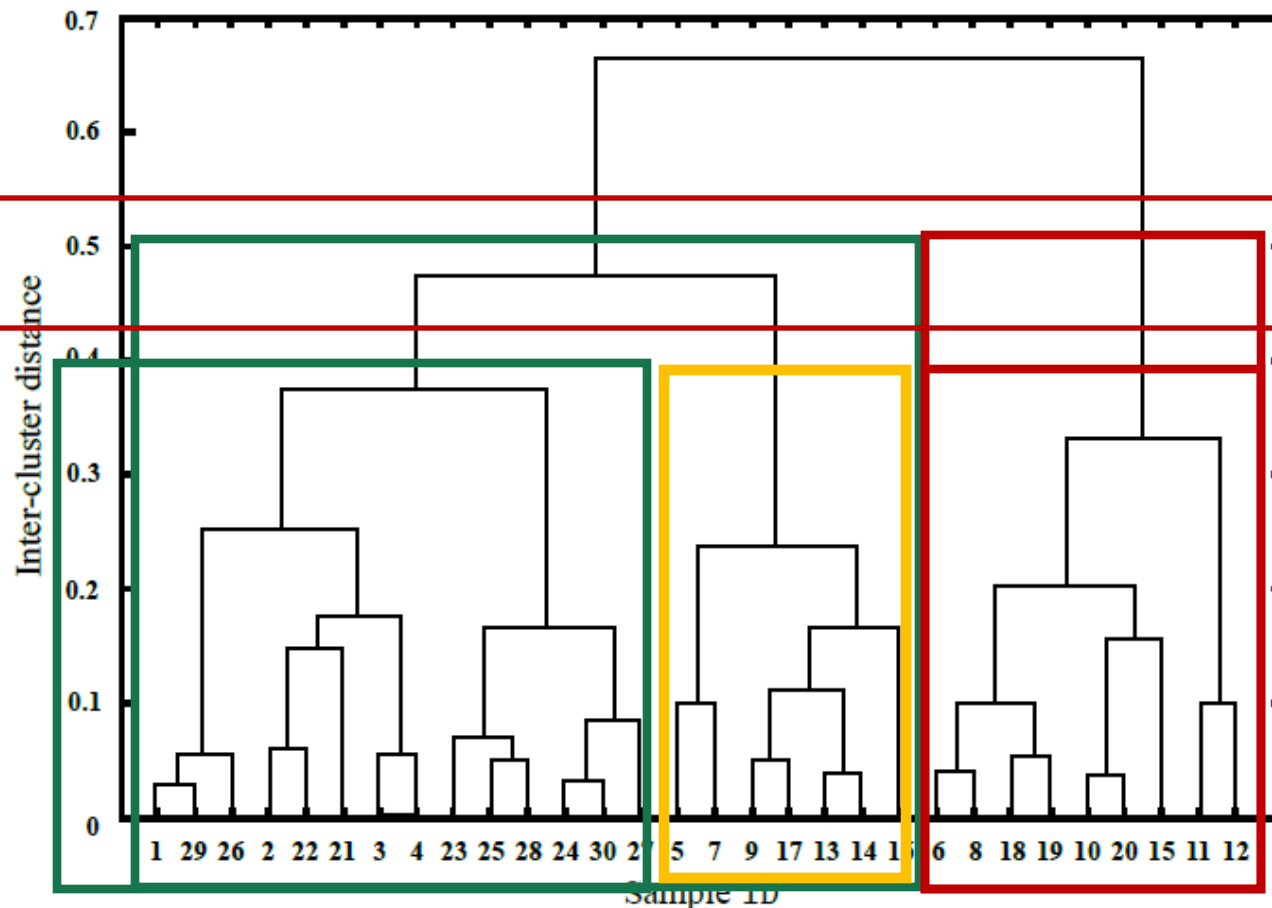
- Clustering Problem
- Performance Measure
- Distance Calculation
- Prototype Clustering
- Density Clustering
- **Hierarchical Clustering**
- Probabilistic clustering

Hierarchical Clustering

- ❑ Hierarchical Clustering aims to create a tree-like clustering structure by dividing a data set at different layers. The hierarchy of clusters can be formed by taking either a *bottom-up* strategy or a *top-down* strategy.

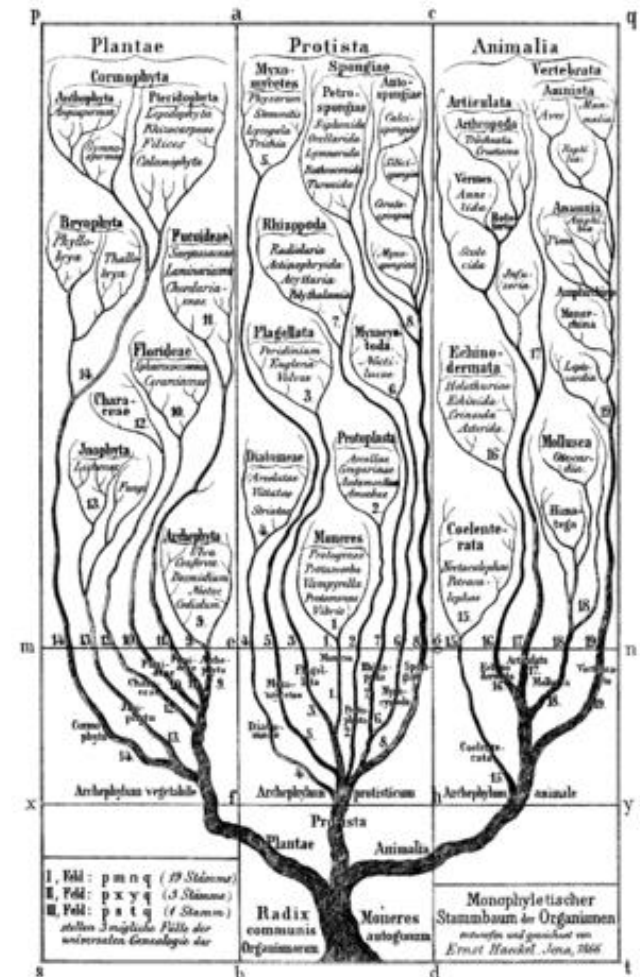
Strengths of Hierarchical Clustering

- ❑ Do not have to assume any particular number of clusters
 - Any desired number of clusters can be obtained by 'cutting' the dendrogram at the proper level



Strengths of Hierarchical Clustering

- They may correspond to meaningful taxonomies
 - Example in biological sciences (e.g., animal kingdom, phylogeny reconstruction, ...)



Hierarchical Clustering

- ❑ AGNES algorithm (bottom-up Hierarchical Clustering)
 - starts by considering each sample in the data set as an initial cluster.
 - Then, in each round, two nearest clusters are merged as a new cluster,
 - This process repeats until the number of clusters meets the pre-specified value.

- ❑ We define the distances of given clusters C_i and C_j in different forms.

Hierarchical Clustering: distances between clusters

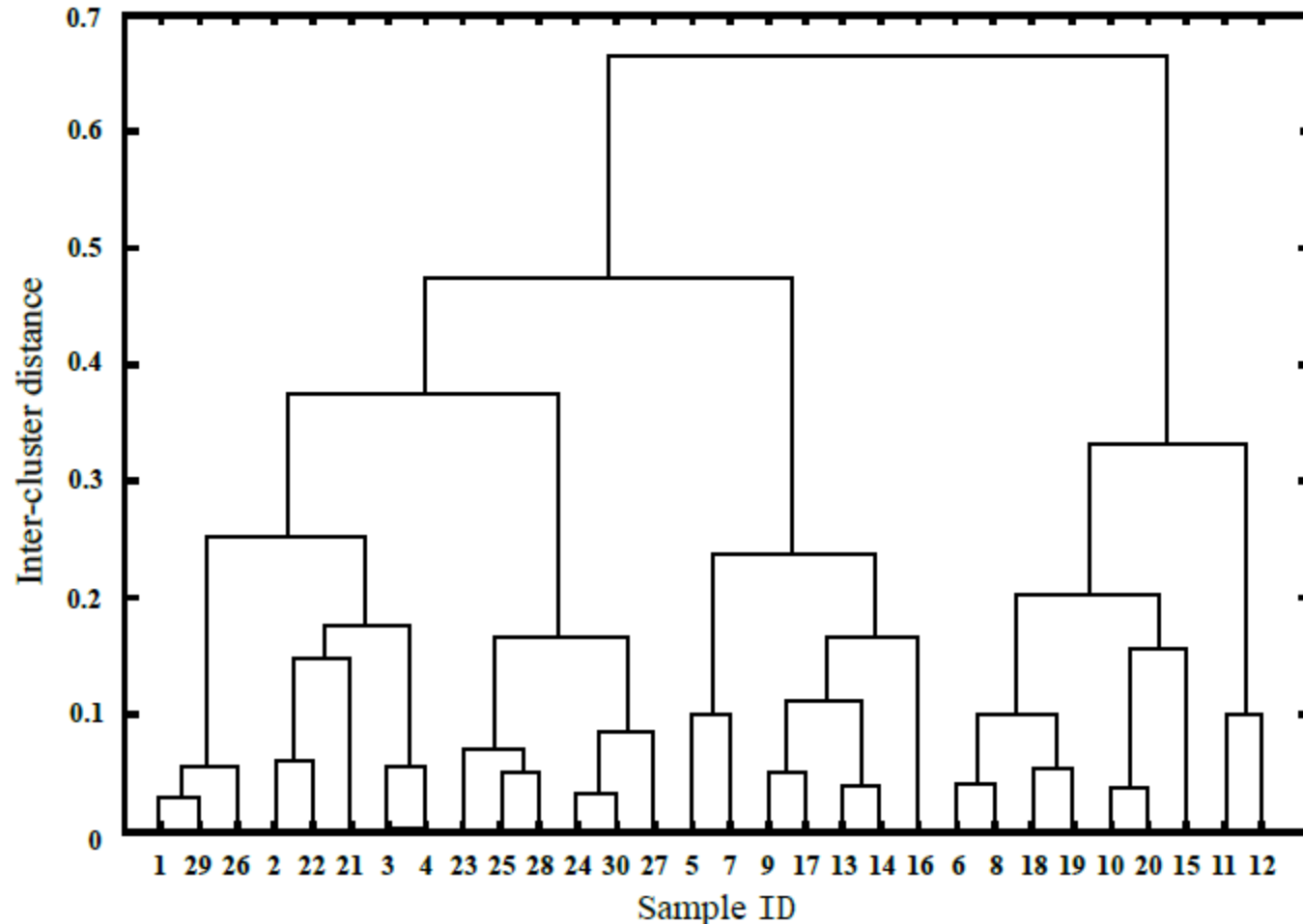
Minimum distance: $d_{min}(C_i, C_j) = \min_{x \in C_i, z \in C_j} dist(x, z)$

Maximum distance: $d_{max}(C_i, C_j) = \max_{x \in C_i, z \in C_j} dist(x, z)$

Average distance: $d_{avg}(C_i, C_j) = \frac{1}{|C_i||C_j|} \sum_{x \in C_i} \sum_{z \in C_j} dist(x, z)$

Hierarchical Clustering – dendrogram

□ The dendrogram of AGNES:



Hierarchical Clustering – AGNES

□ Pseudocode for AGNES:

Algorithm 9.5 AGNES.

Input: Data set $D = \{x_1, x_2, \dots, x_m\}$;
Cluster distance metric function d ; $\longrightarrow$ can be $d_{\min}$, $d_{\max}$, d_{avg}
Number of clusters k .

Process:

```
1: for  $j = 1, 2, \dots, m$  do
2:    $C_j = \{x_j\}$ ;  $\longrightarrow$  Starts with singleton clusters
3: end for
4: for  $i = 1, 2, \dots, m$  do
5:   for  $j = i + 1, \dots, m$  do
6:      $M(i, j) = d(C_i, C_j)$ ;  $\longrightarrow$  Matrix of point to point
7:      $M(j, i) = M(i, j)$ ;      distances (symmetric)
8:   end for
9: end for
10: Set the current number of clusters:  $q = m$ ;  $\longrightarrow$  actual number of tree levels
11: while  $q > k$  do  $\longrightarrow$  Until the number of levels is reached
12:   Find two clusters  $C_{i^*}$  and  $C_{j^*}$  that have the shortest distance;
13:   Merge  $C_{i^*}$  and  $C_{j^*}$ :  $C_{i^*} = C_{i^*} \cup C_{j^*}$ ;
14:   for  $j = j^* + 1, j^* + 2, \dots, q$  do
15:     Change the index of  $C_j$  to  $C_{j-1}$ ;  $\longrightarrow$  Shifts clusters indexes
16:   end for                                     so that cluster  $j$  will be
17:   Delete the  $j^*$ th row and  $j^*$ th column of the distance matrix  $M$ ;      forgotten
18:   for  $j = 1, 2, \dots, q - 1$  do
19:      $M(i^*, j) = d(C_{i^*}, C_j)$ ;  $\longrightarrow$  Updates the distance matrix
20:      $M(j, i^*) = M(i^*, j)$ ;      considering the presence of the
21:   end for                                     new cluster  $i^*$  and all other clusters  $j$ 
22:    $q = q - 1$ .
23: end while
```

Output: Clusters $\mathcal{C} = \{C_1, C_2, \dots, C_k\}$.

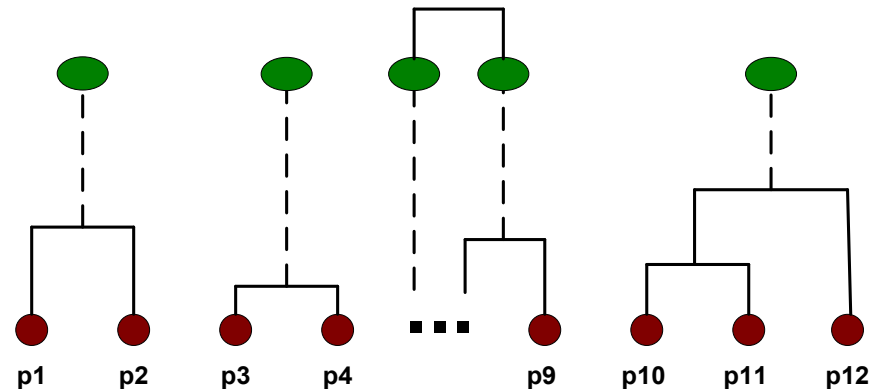
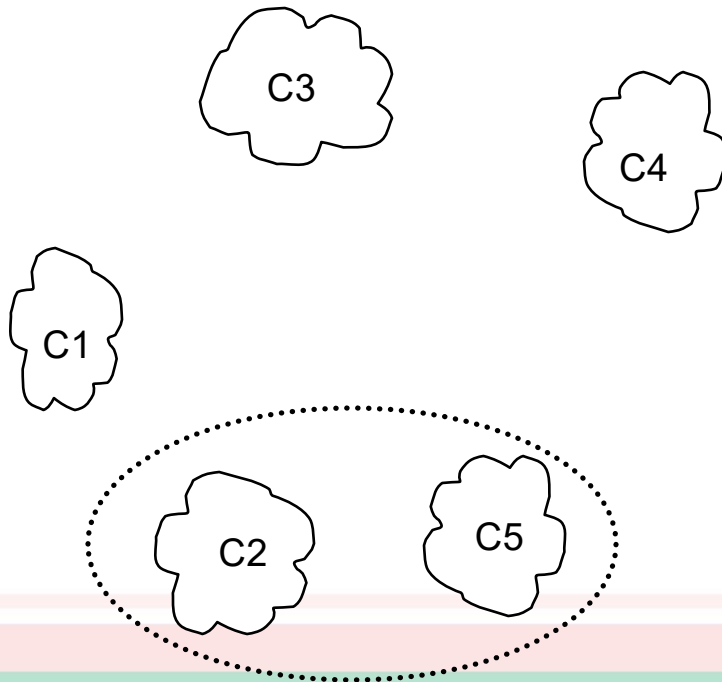
Example: distance matrix update

□ When two closest clusters (C_2 and C_5) are merged we need to update the proximity matrix:

- Keep only one row/column (C_2)
- Delete the other one (C_5)
- Update the distances between the other clusters and the new one (C_2^*)

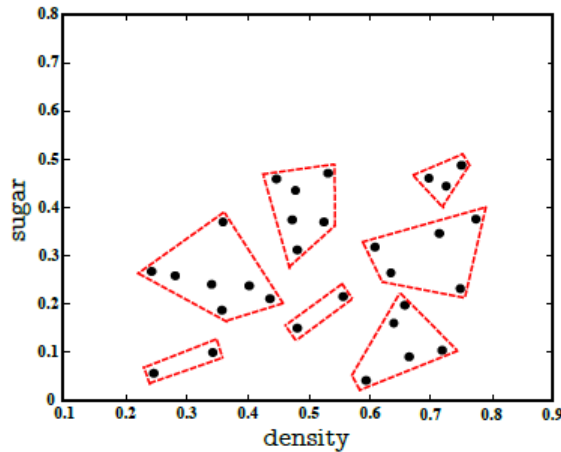
	C1	C2	C3	C4	C5
C1					
C2					
C3					
C4					
C5					

Proximity Matrix

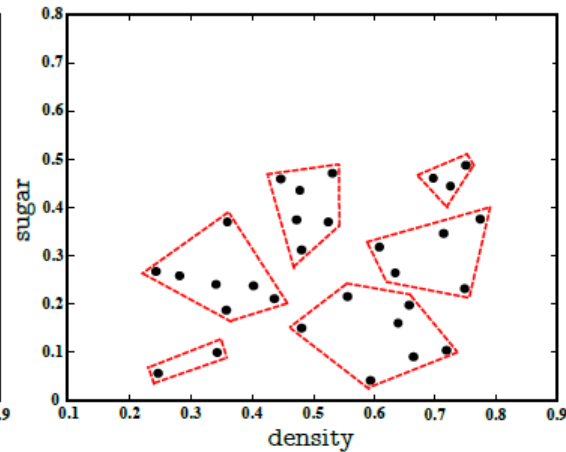


Hierarchical Clustering

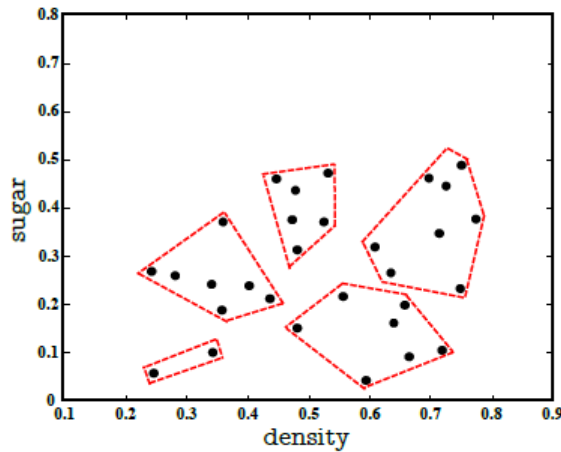
□ AGNES Clustering results :



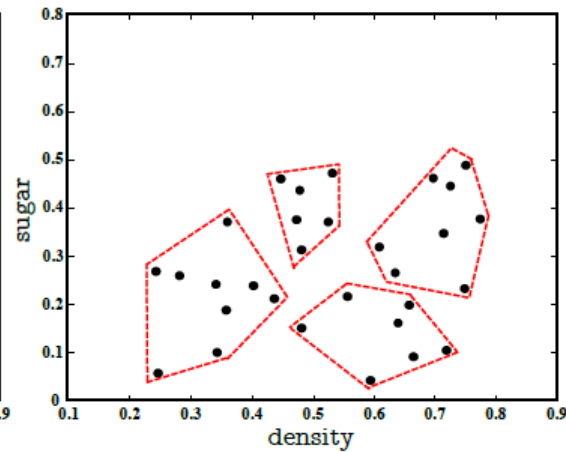
(a) $k = 7$.



(b) $k = 6$.



(c) $k = 5$.



(d) $k = 4$.

Outlines

- Clustering Problem
- Performance Measure
- Distance Calculation
- Prototype Clustering
- Density Clustering
- Hierarchical Clustering
- **Probabilistic clustering (for the course of 9 credits)**

Prototype Clustering – Mixture-of-Gaussian Clustering

Unlike k -means and LVQ, Mixture-of-Gaussian clustering does not use prototype vectors but probabilistic models to represent clustering structures.

□ Definition of multivariate Gaussian distribution

For a random vector x in an n -dimensional sample space $\mathcal{X}$,

$$p(x) = \frac{1}{(2\pi)^{\frac{n}{2}} |\Sigma|^{\frac{1}{2}}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

formula multivariata

where μ is an n -dimensional mean vector and Σ is an $n \times n$ covariance matrix. We write the probability density function as $p(x|\mu, \Sigma)$

Computation of the covariance matrix Σ

- Data Matrix X (m objects, n features)

$$\mathbf{X} = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{1m} & \cdots & x_{mn} \end{bmatrix}$$

- Covariance matrix Σ ($n \times n$)

- The σ_{lj} element is the covariance between feature l and j over all objects: it measures how strongly feature l and j vary together in the m objects.
If $l=j$ it is the variance of feature l .

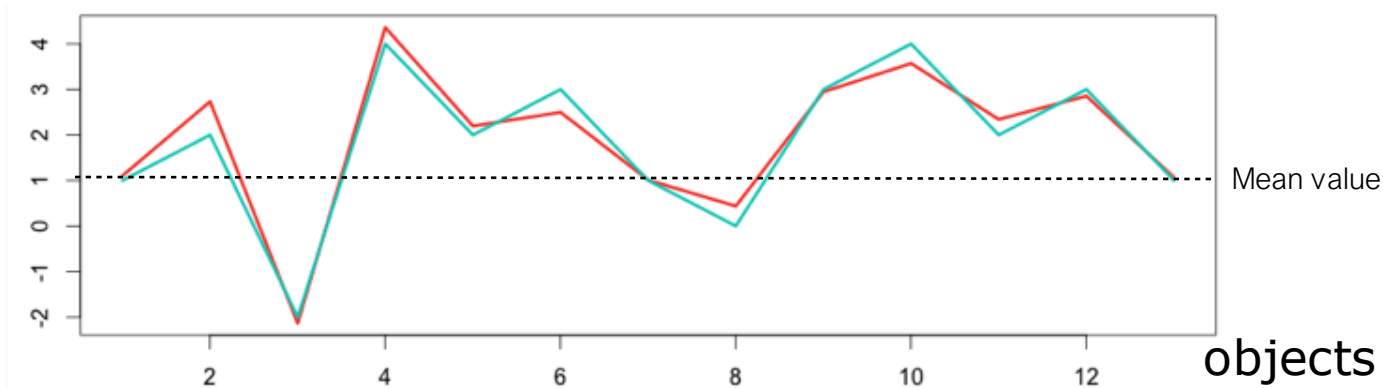
$$\Sigma = \begin{bmatrix} \sigma_{11} & \cdots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{1n} & \cdots & \sigma_{nn} \end{bmatrix}$$

$$\sigma_{lj} = \frac{1}{m} \sum_{k=1}^m \overset{1^\circ \text{ scarto}}{(x_{kl} - \bar{x}_l)} \overset{2^\circ \text{ scarto}}{(x_{kj} - \bar{x}_j)}$$

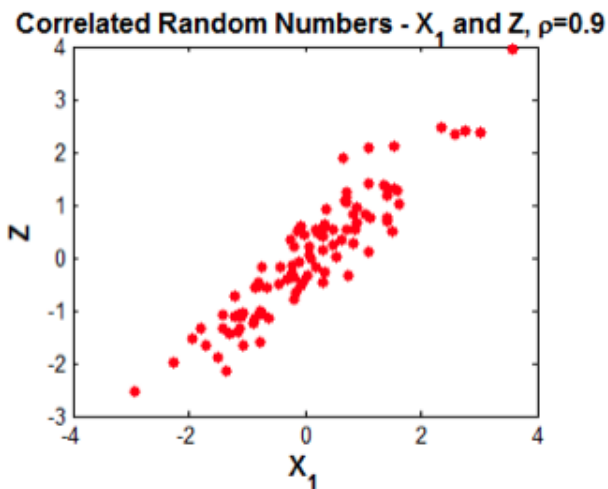
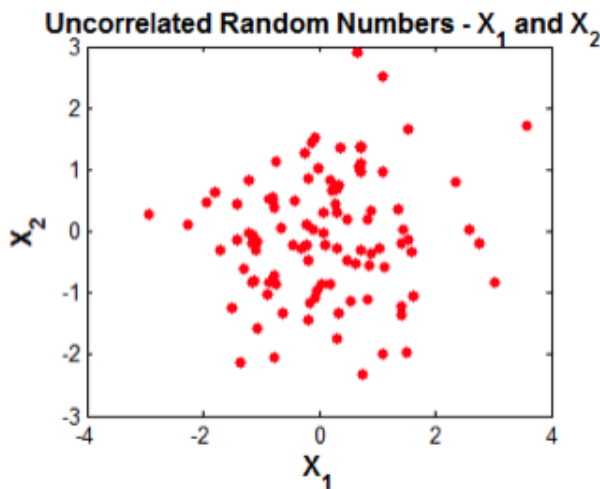
mean value of feature l mean value of feature j

Showing the correlation between two features

- Positively correlated features (point values in red and green) lines)



- Scatter plots:



coefficiente di covsrinza tra le due feature

$$\rho = \frac{\sigma_{12}}{\sqrt{\sigma_{11} \cdot \sigma_{22}}}$$

Prototype Clustering – Mixture-of-Gaussian Clustering

- Definition of the Mixture-of-Gaussian distribution

$$p_M(x) = \sum_{i=1}^k \alpha_i p(x | \mu_i, \Sigma_i)$$

i-esima distribuzione gaussiana

- It consists of k mixture components:
- each component corresponds to a Gaussian distribution.
- μ_i and Σ_i are the parameters of the i -th mixture component;
- $\alpha_i > 0$ are the corresponding *mixture coefficients*, where
 $\sum_{i=1}^k \alpha_i = 1$ (the mixture is finite).

A Two-Cluster Mixture Model

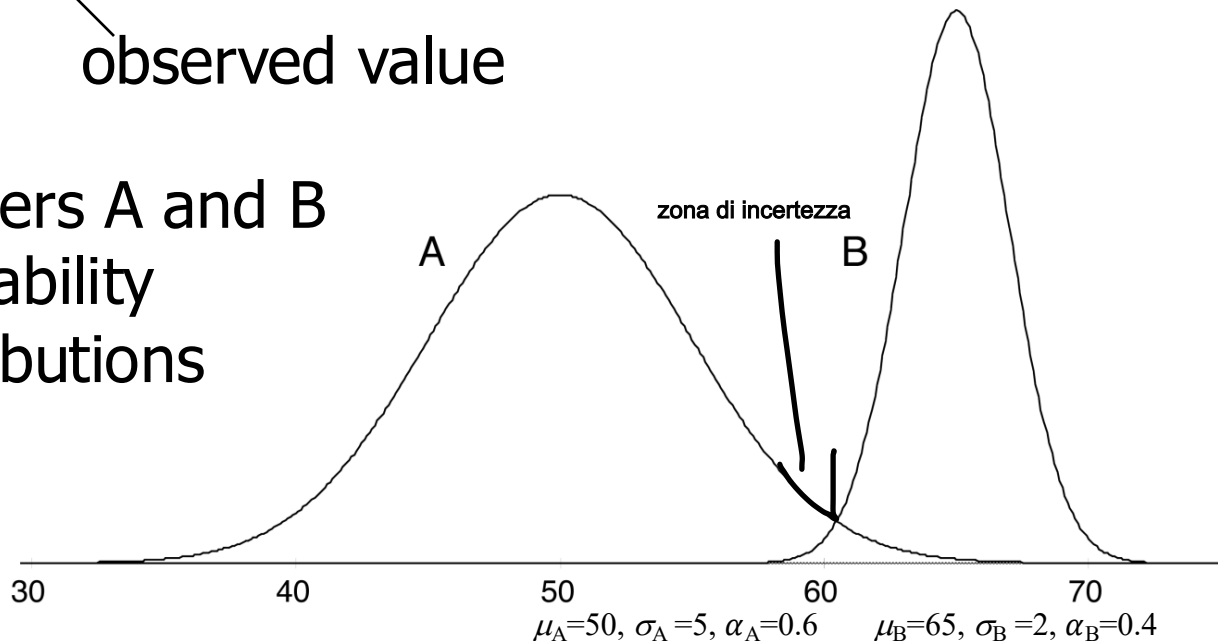
data

A	51	B	62	B	64	A	48	A	39	A	51
A	43	A	47	A	51	B	64	B	62	A	48
B	62	A	52	A	52	A	51	B	64	B	64
B	64	B	64	B	62	B	63	A	52	A	42
A	45	A	51	A	49	A	43	B	63	A	48
A	42	B	65	A	48	B	65	B	64	A	41
A	46	A	48	B	62	B	66	A	48		
A	45	A	49	A	43	B	65	B	64		
A	45	A	46	A	40	A	46	A	48		

model

cluster observed value

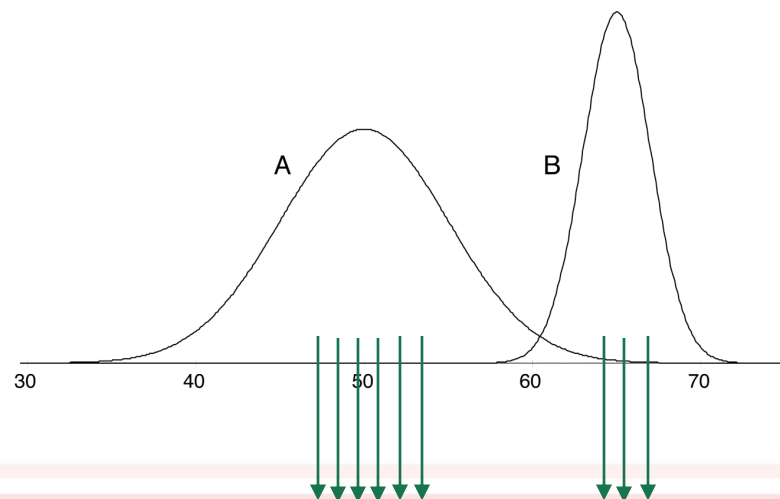
Clusters A and B
probability
distributions



Prototype Clustering – Mixture-of-Gaussian Clustering

- Suppose that the samples are generated from a Mixture-of-Gaussian distribution:
 1. Firstly select the Gaussian mixture components using the prior distribution defined by $\alpha_1, \alpha_2, \dots, \alpha_k$, where α_i is the probability of selecting the i -th mixture component
 2. Then, generate samples by sampling from the probability density functions of the selected mixture components.

pedice j -> cluster
pedice i -> oggetto



Here samples will be more probable

Prototype Clustering – Mixture-of-Gaussian Clustering

- Optimization of the model parameters: maximum the likelihood

$$p_M(x) = \sum_{i=1}^k \alpha_i p(x|\mu_i, \Sigma_i)$$

$$LL(D) = \ln \left(\prod_{j=1}^m p_M(x_j) \right)$$

$$= \sum_{j=1}^m \ln \left(\sum_{i=1}^k \alpha_i \cdot p(x_j|\mu_i, \Sigma_i) \right)$$

Let: $\frac{\partial LL(D)}{\partial \mu_i} = 0$  $\mu_i = \frac{\sum_{j=1}^m \gamma_{ji} x_j}{\sum_{j=1}^m \gamma_{ji}}$

where:

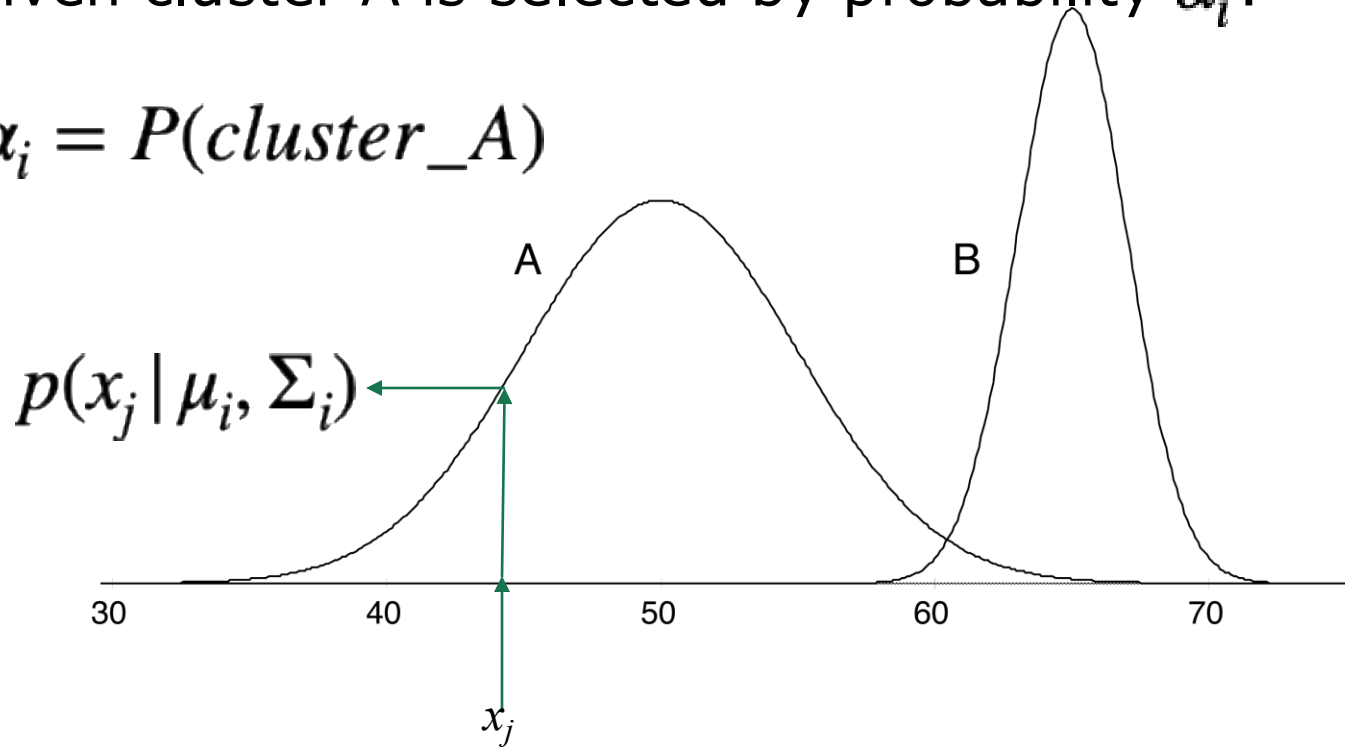
$$\gamma_{ji} = p_{\mathcal{M}}(z_j = i | x_j) (1 \leq i \leq k);$$

$$\gamma_{ji} = \frac{\alpha_i \cdot p(x_j | \mu_i, \Sigma_i)}{\sum_{l=1}^k \alpha_l \cdot p(x_j | \mu_l, \Sigma_l)}$$

Example

- Given cluster A is selected by probability α_i :

$$\alpha_i = P(\text{cluster_A})$$



$$\gamma_{ji} = \frac{\alpha_i \cdot p(x_j | \mu_i, \Sigma_i)}{\sum_{l=1}^k \alpha_l \cdot p(x_j | \mu_l, \Sigma_l)}$$

$$\mu_i = \frac{\sum_{j=1}^m \gamma_{ji} \cdot x_j}{\sum_{j=1}^m \gamma_{ji}}$$

Mixture-of-Gaussian Clustering-Optimization (Continued)

Let:

$$\frac{\partial LL(D)}{\partial \Sigma_i} = 0 \quad \longrightarrow \quad \Sigma_i = \frac{\sum_{j=1}^m \gamma_{ji} (x_j - \mu_i)(x_j - \mu_i)^T}{\sum_{j=1}^m \gamma_{ji}}$$

Lagrange multiplier: $\longrightarrow \alpha_i = \frac{1}{m} \sum_{j=1}^m \gamma_{ji}$

Mixture-of-Gaussian Clustering

□ Pseudocode:

Algorithm 9.3 Mixture-of-Gaussian Clustering.

Input: Data set $D = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m\}$;

Number of Gaussian mixture components k .

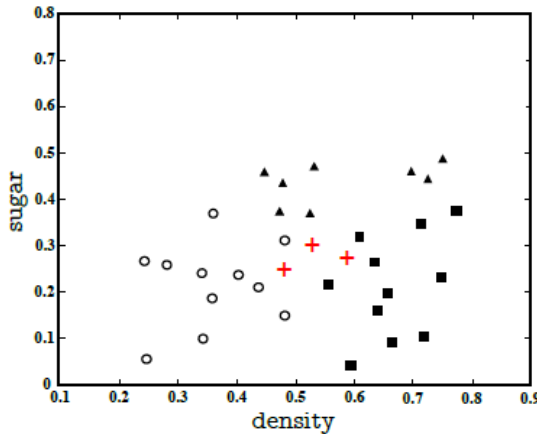
Process:

- 1: Initialize the parameters $\{(\alpha_i, \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) \mid 1 \leq i \leq k\}$ of the Mixture-of-Gaussian distribution;
- 2: **repeat**
- 3: **for** $j = 1, 2, \dots, m$ **do**
- 4: According to (9.30), compute the posterior probabilities that $\mathbf{x}_j$ is generated by each Gaussian mixture component, i.e., $\gamma_{ji} = p_{\mathcal{M}}(z_j = i \mid \mathbf{x}_j) (1 \leq i \leq k)$; $p_{\mathcal{M}}(z_j = i \mid \mathbf{x}_j) = \frac{P(z_j = i) \cdot p_{\mathcal{M}}(\mathbf{x}_j \mid z_j = i)}{p_{\mathcal{M}}(\mathbf{x}_j)}$
- 5: **end for**
- 6: **for** $i = 1, 2, \dots, k$ **do**
- 7: Compute the updated mean vector: $\boldsymbol{\mu}'_i = \frac{\sum_{j=1}^m \gamma_{ji} \mathbf{x}_j}{\sum_{j=1}^m \gamma_{ji}}$; $= \frac{\alpha_i \cdot p(\mathbf{x}_j \mid \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)}{\sum_{l=1}^k \alpha_l \cdot p(\mathbf{x}_j \mid \boldsymbol{\mu}_l, \boldsymbol{\Sigma}_l)}$
- 8: Compute the updated covariance matrix: $\boldsymbol{\Sigma}'_i = \frac{\sum_{j=1}^m \gamma_{ji} (\mathbf{x}_j - \boldsymbol{\mu}'_i)(\mathbf{x}_j - \boldsymbol{\mu}'_i)^T}{\sum_{j=1}^m \gamma_{ji}}$;
- 9: Compute the updated mixture coefficients: $\alpha'_i = \frac{1}{m} \sum_{j=1}^m \gamma_{ji}$;
- 10: **end for**
- 11: Update the model parameters $\{(\alpha_i, \boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i) \mid 1 \leq i \leq k\}$ to $\{(\alpha'_i, \boldsymbol{\mu}'_i, \boldsymbol{\Sigma}'_i) \mid 1 \leq i \leq k\}$;
- 12: **until** The termination condition is met
- 13: $C_i = \emptyset (1 \leq i \leq k)$;
- 14: **for** $j = 1, 2, \dots, m$ **do**
- 15: Determine the cluster label λ_j of $\mathbf{x}_j$ according to (9.31); $\longrightarrow \lambda_j = \arg \max_{i \in \{1, 2, \dots, k\}} \gamma_{ji}$.
- 16: Move $\mathbf{x}_j$ to the corresponding cluster: $C_{\lambda_j} = C_{\lambda_j} \cup \{\mathbf{x}_j\}$.
- 17: **end for**

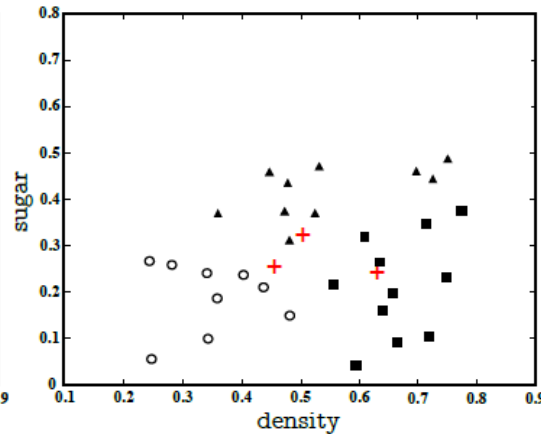
Output: Clusters $\mathcal{C} = \{C_1, C_2, \dots, C_k\}$.

Mixture-of-Gaussian Clustering

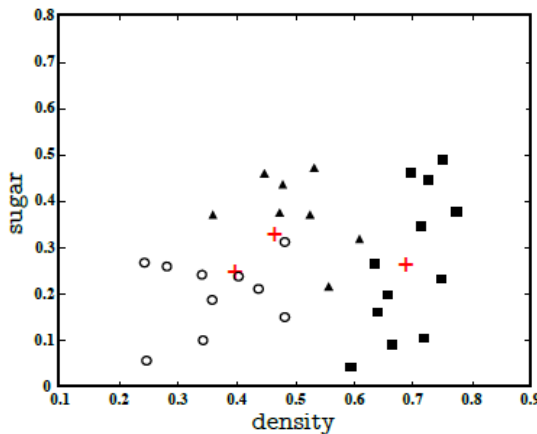
□ Clustering results



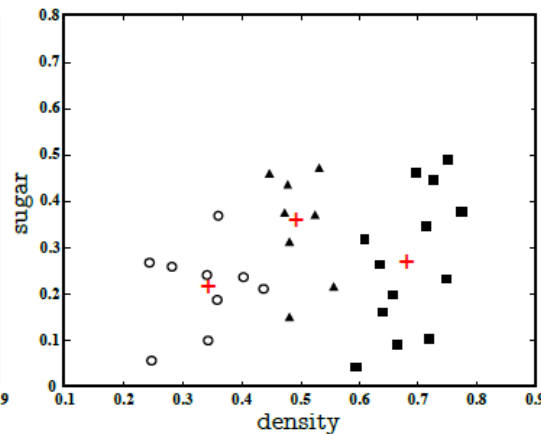
(a) After 5 iterations.



(b) After 10 iterations.



(c) After 20 iterations.



(d) After 50 iterations.

la convergenza rispetto al k-means è più lenta ma
allo stesso tempo più potente

Summary

- ❑ **Clustering Problem**
- ❑ **Performance Measure**
- ❑ **Distance Calculation**
- ❑ **Prototype Clustering**
- ❑ **Density Clustering**
- ❑ **Hierarchical Clustering**
- ❑ **Probabilistic clustering**

Thanks!
